KỸ THUẬT SỐ

Đánh giá

- **Điểm quá trình: 50%**
  - Chuyên cần
  - Kiểm tra
  - Tiểu luận
- **Điểm cuối kỳ: 50%**

Tài liệu tham khảo

- Nguyễn Trường Duy, Giáo trình Kỹ thuật số, Đại học Quốc gia Tp. HCM, Năm 2019.
- Ronald J. Tocci - Neal S. Widmer – Gregory L. Moss, Digital Systems: Principles and Applications, 12th Ed. Prentice Hall, 2017.
- Thomas L. Floyd, Digital Fundamental, 11th Ed. Prentice Hall, 2015.

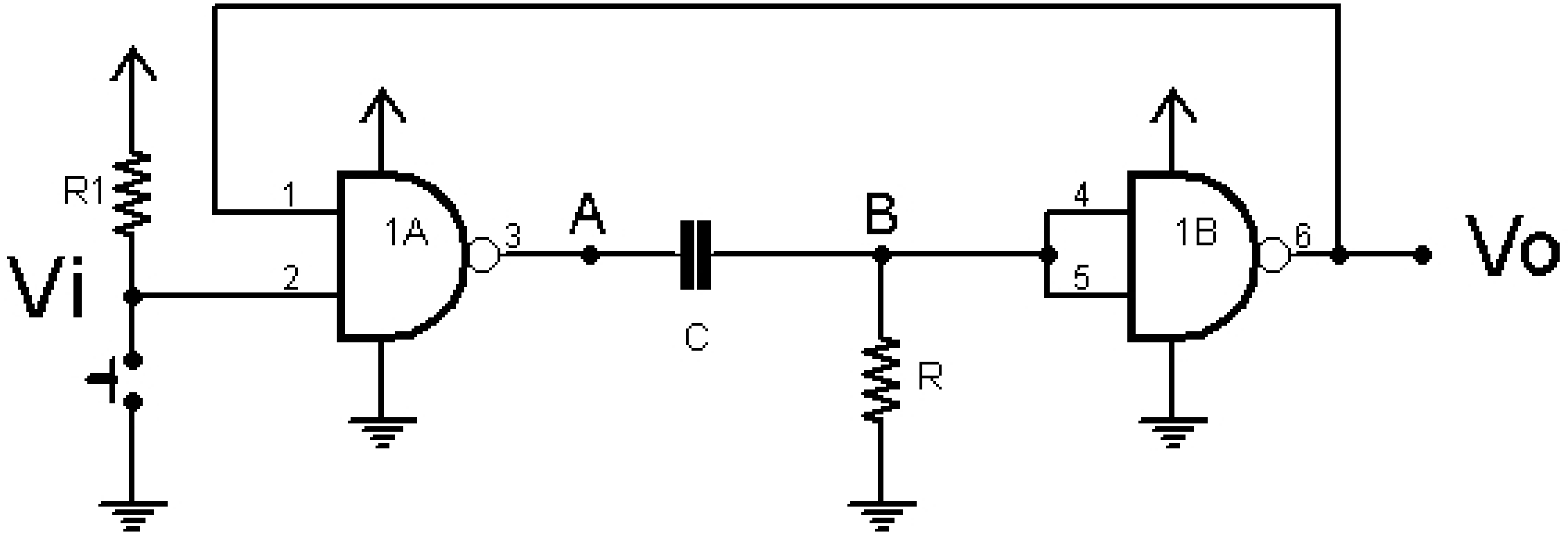
Timer

Nội dung

- Mạch đơn ổn
- Mạch dao động
- IC 555

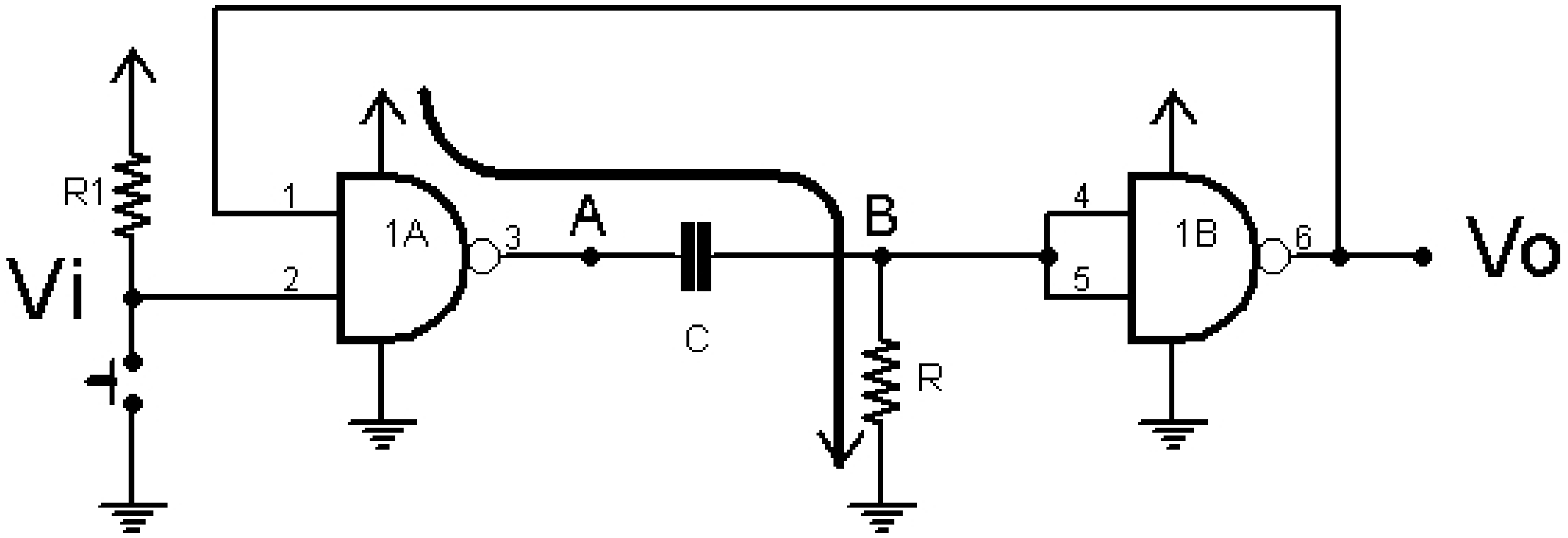
Mạch đơn ổn

- Mạch đơn ổn dùng cổng NAND



Mạch đơn ổn

- Mạch đơn ổn dùng cổng NAND



Mạch đơn ổn

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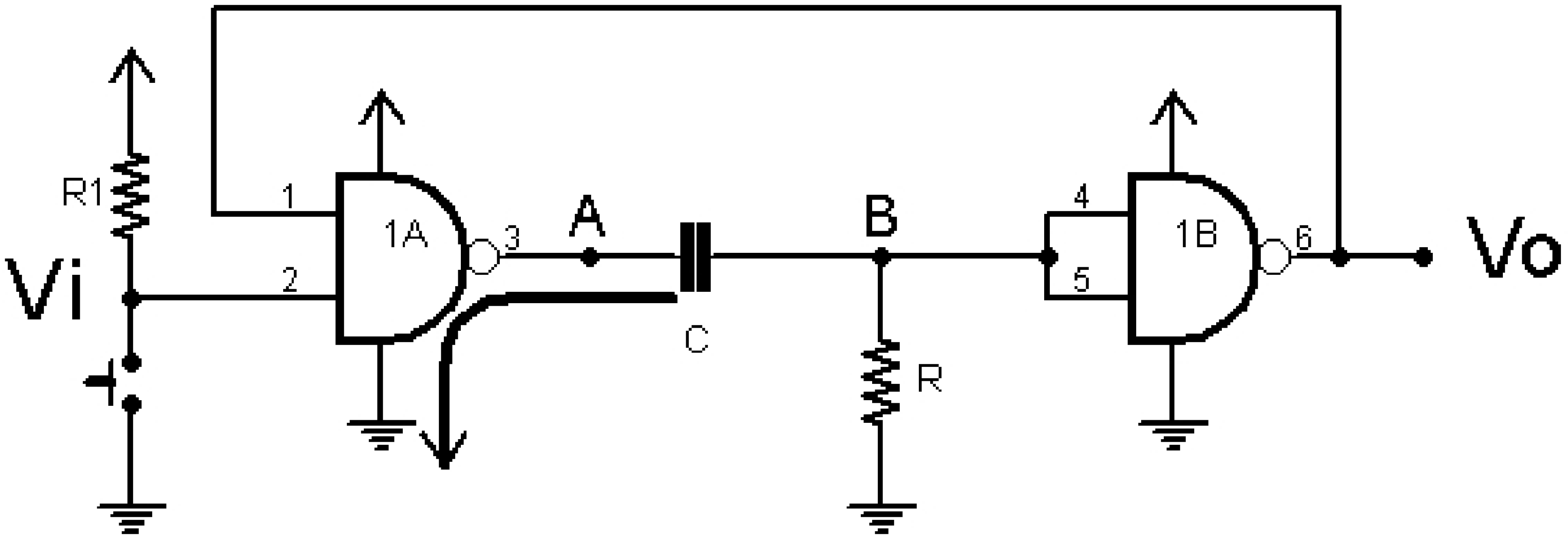
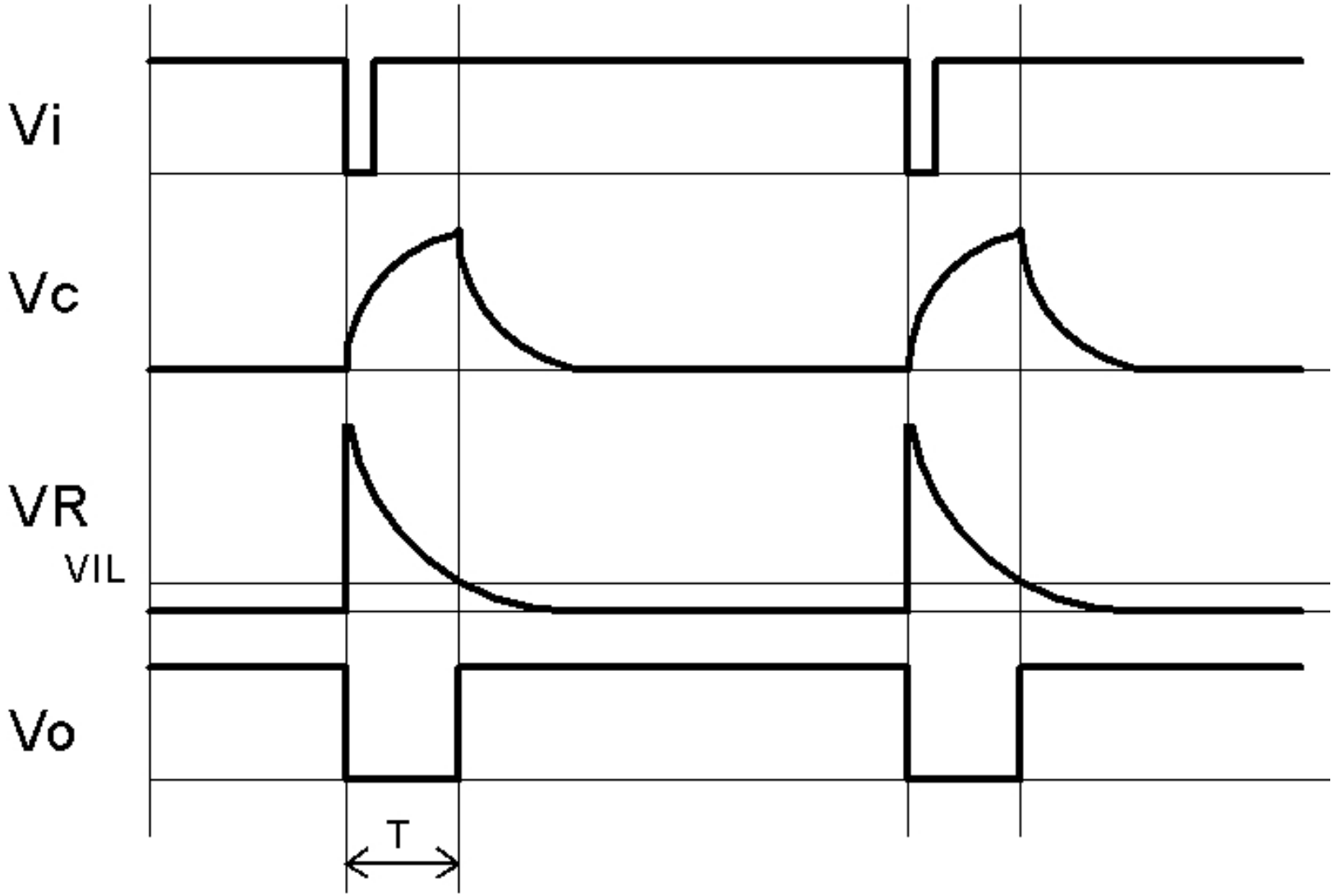


Mạch đơn ổn

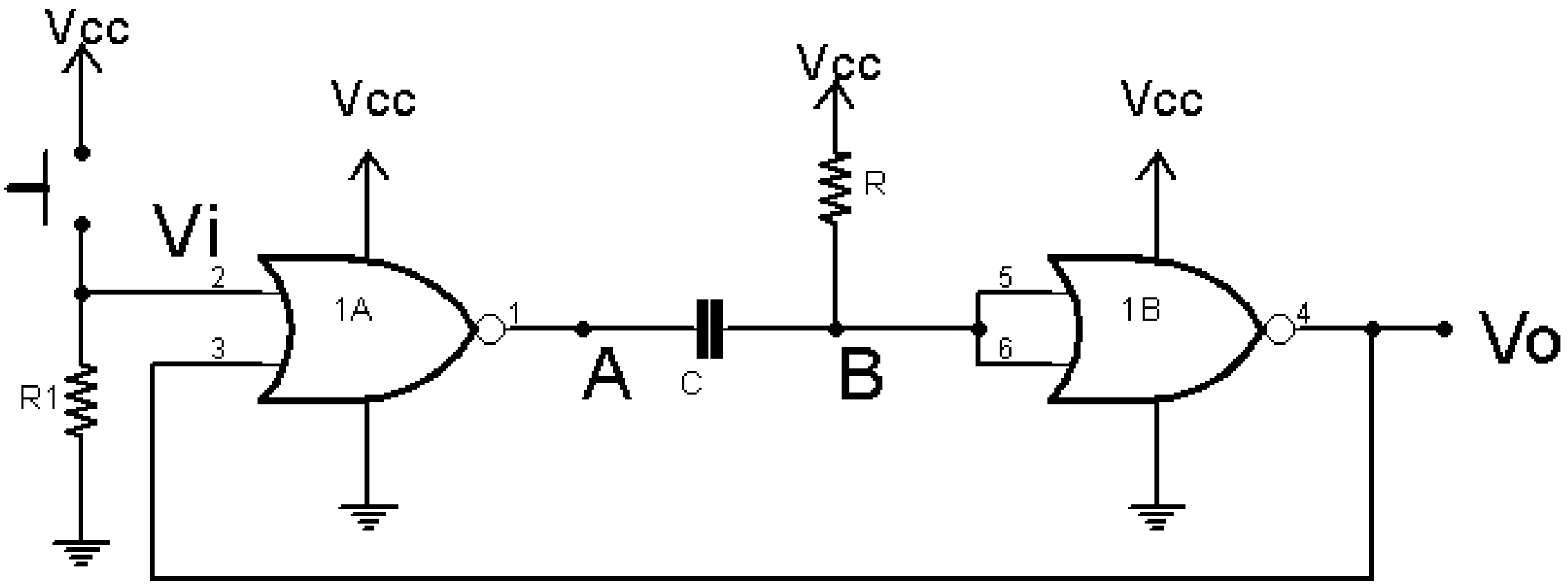
- Mạch đơn ổn dùng cổng NAND



$$T \approx RC\ln 2$$

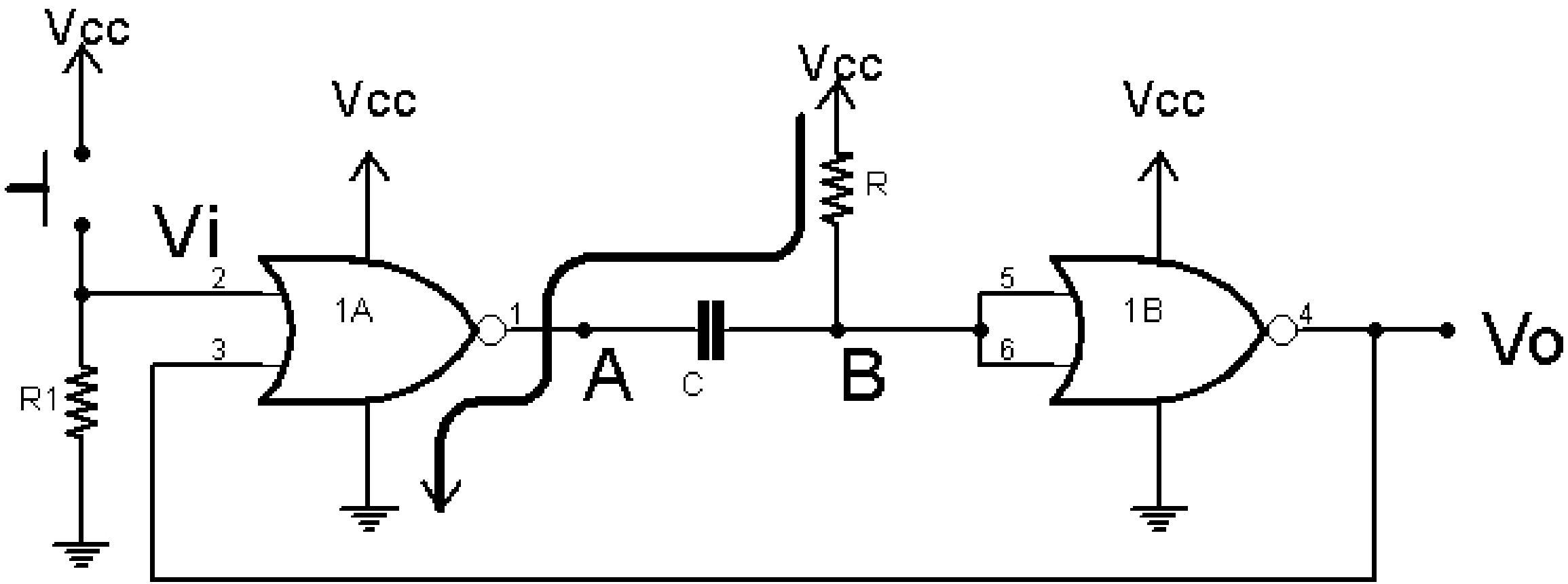
Mạch đơn ổn

- Mạch đơn ổn dùng cổng NOR



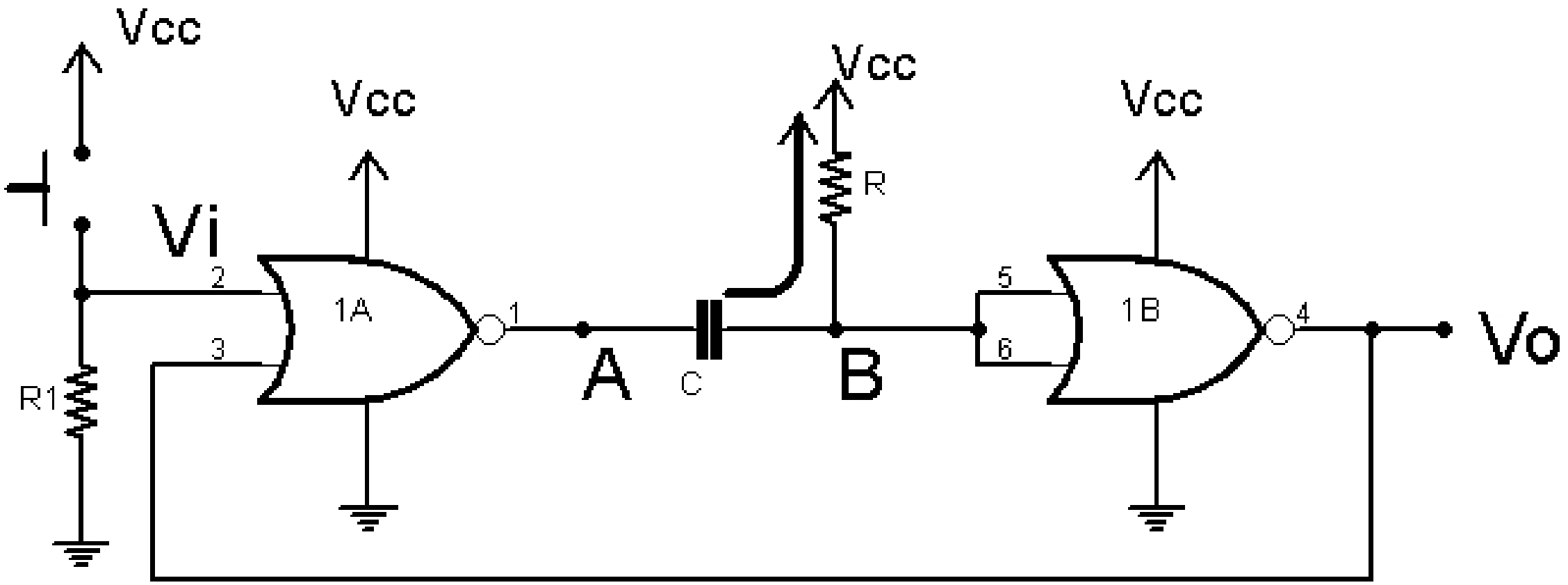
Mạch đơn ổn

- Mạch đơn ổn dùng cổng NOR



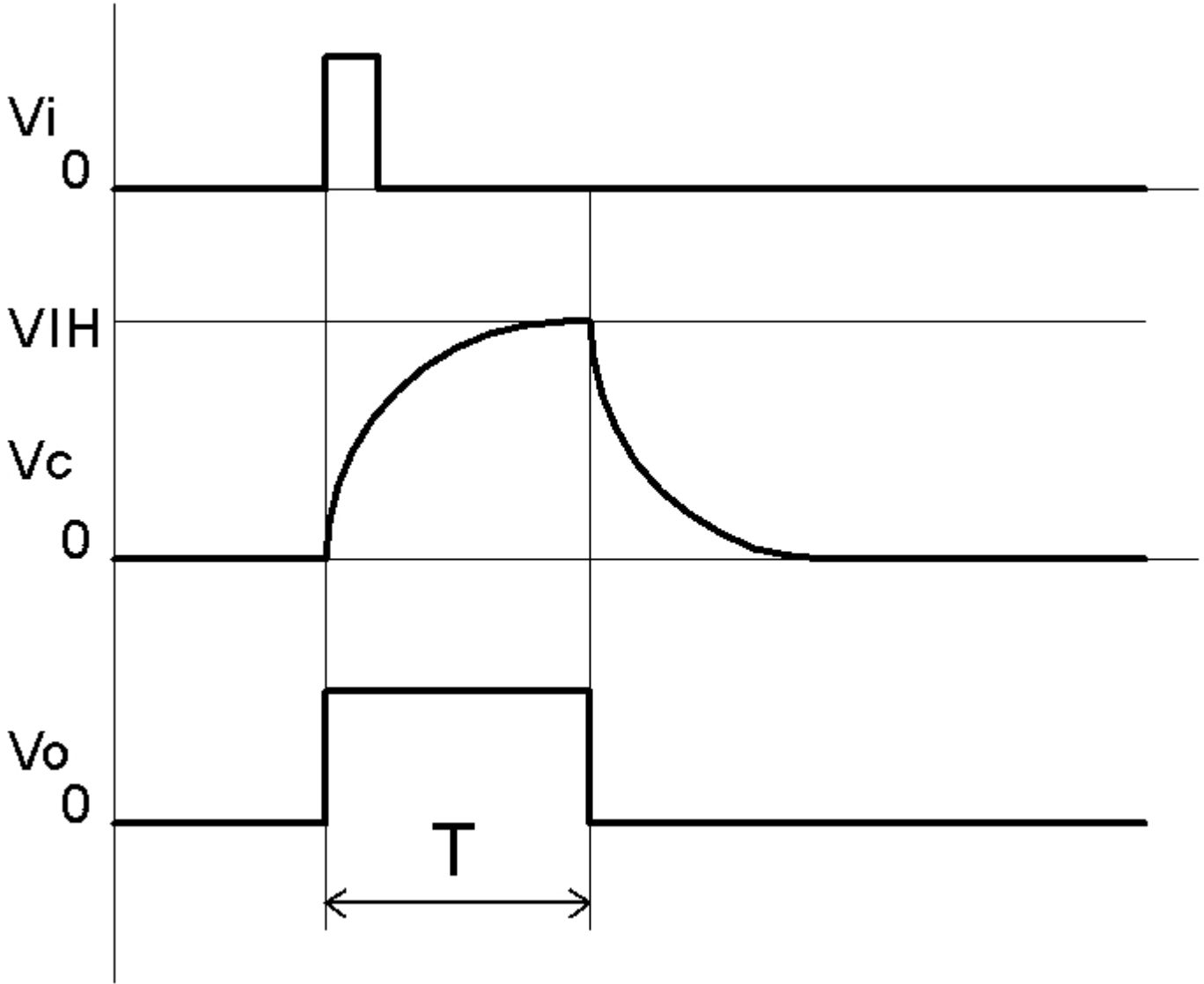
Mạch đơn ổn

- Mạch đơn ổn dùng cổng NOR



Mạch đơn ổn

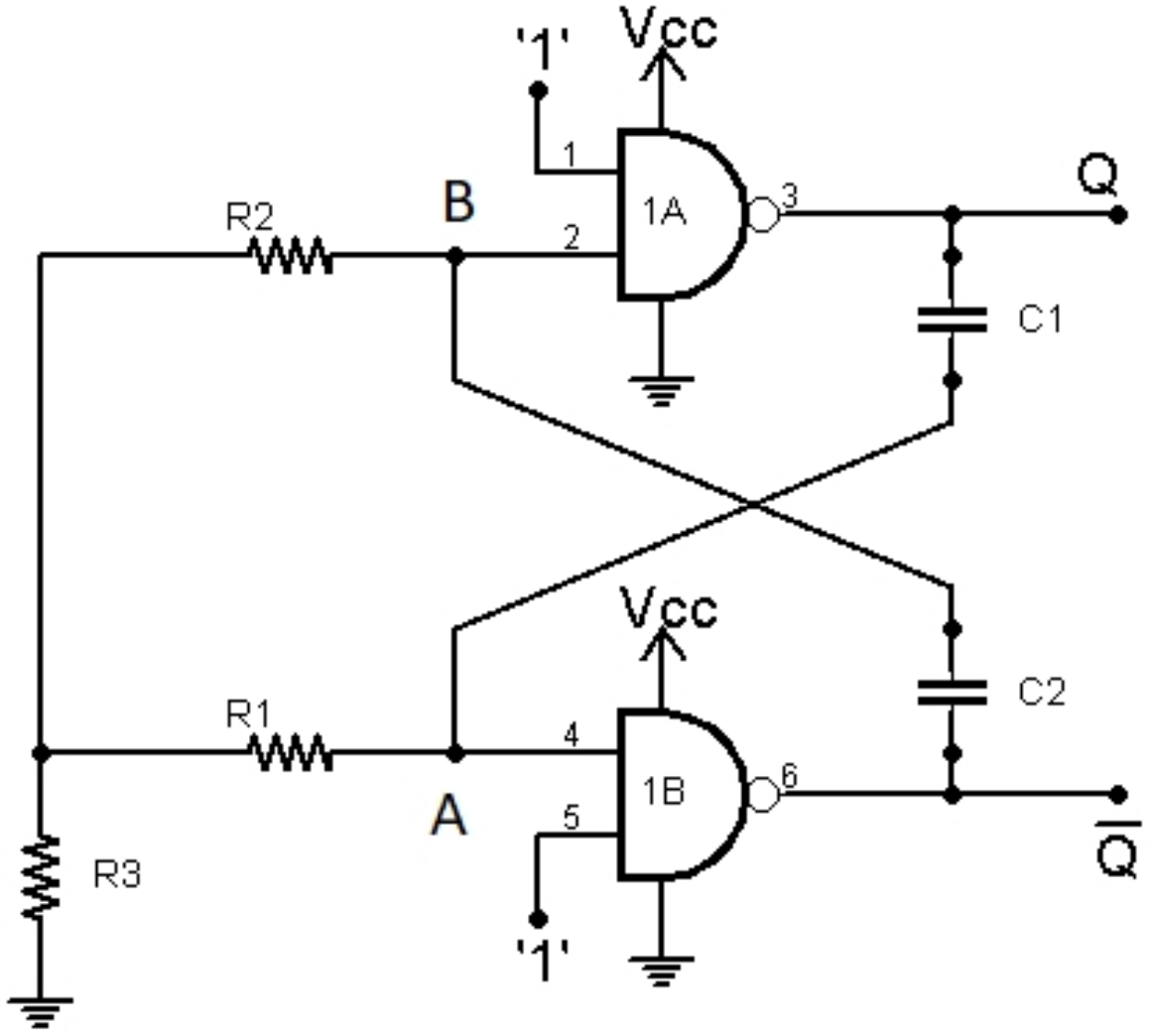
- Mạch đơn ổn dùng cổng NOR



$$T \approx RC\ln 2$$

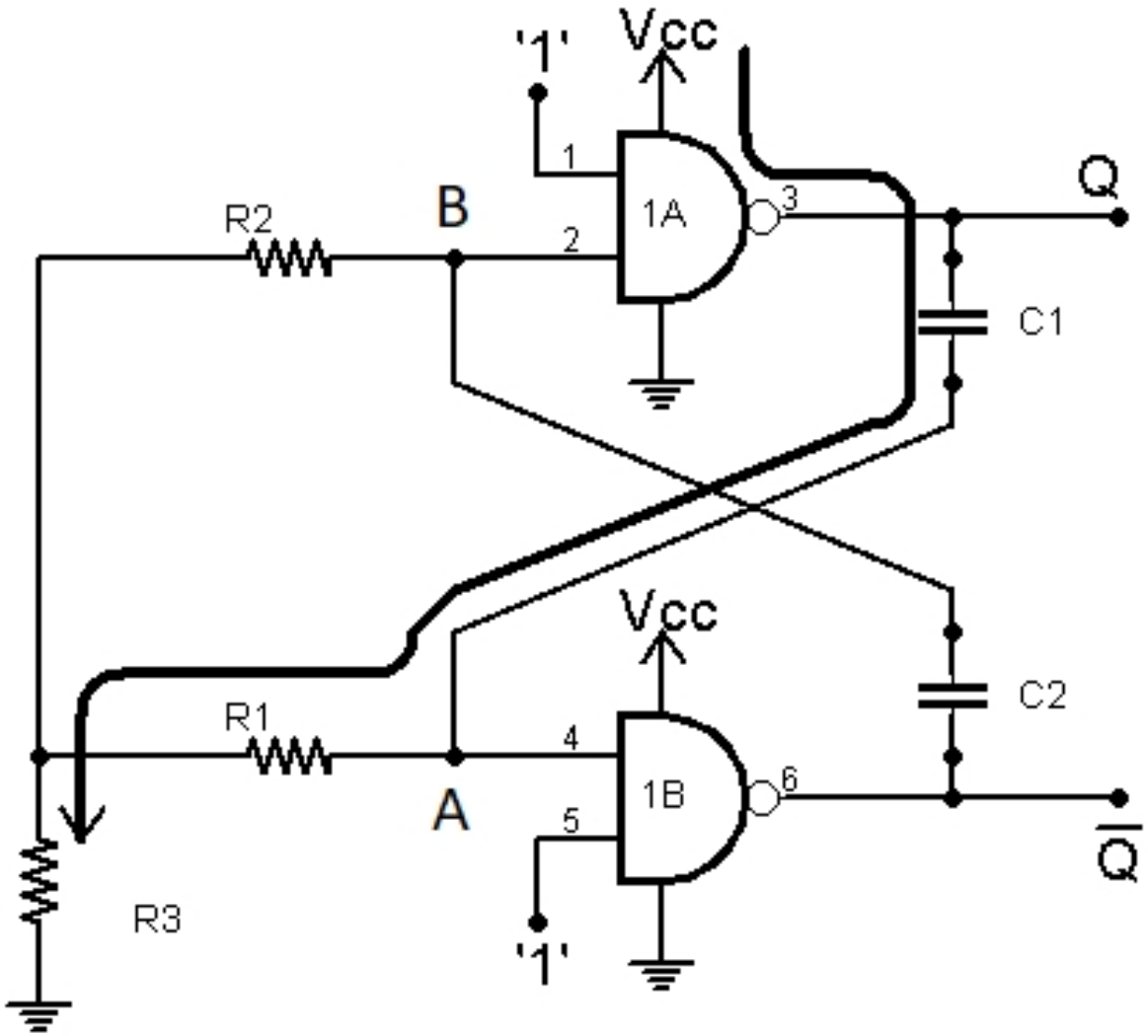
Mạch dao động

- Mạch dao động dùng cổng NAND



Mạch dao động

- Mạch dao động dùng cổng NAND



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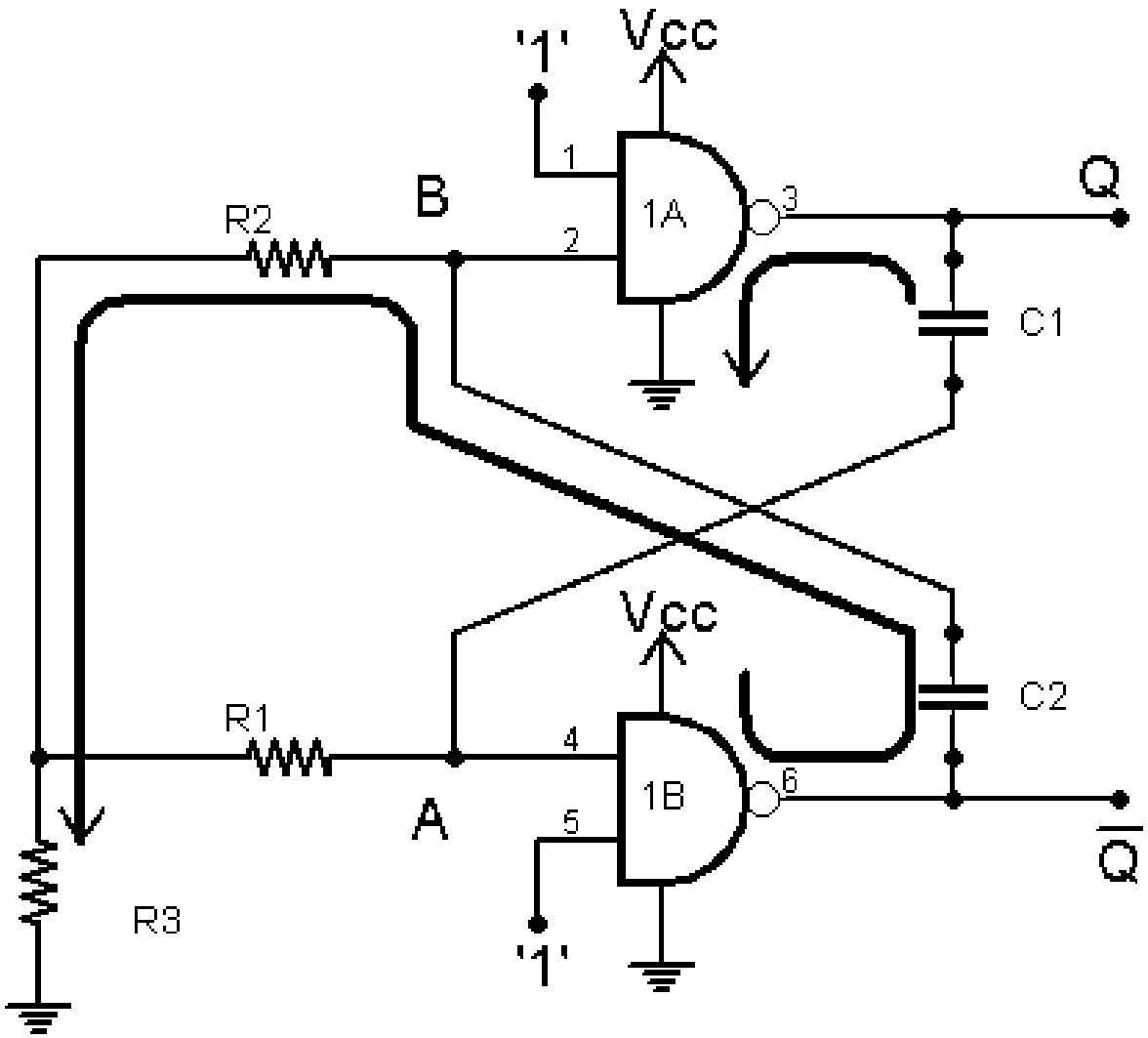

Mạch dao động

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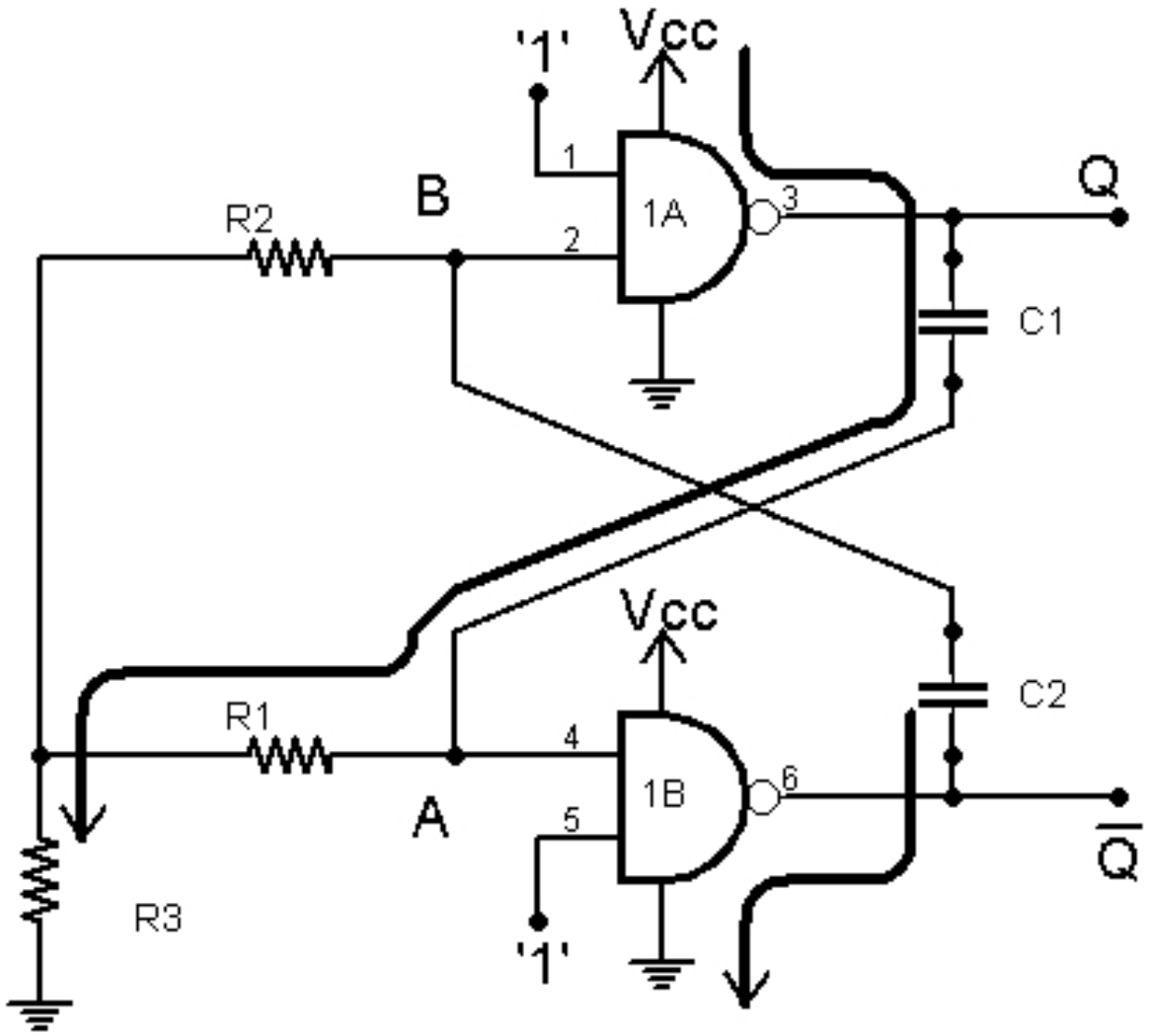
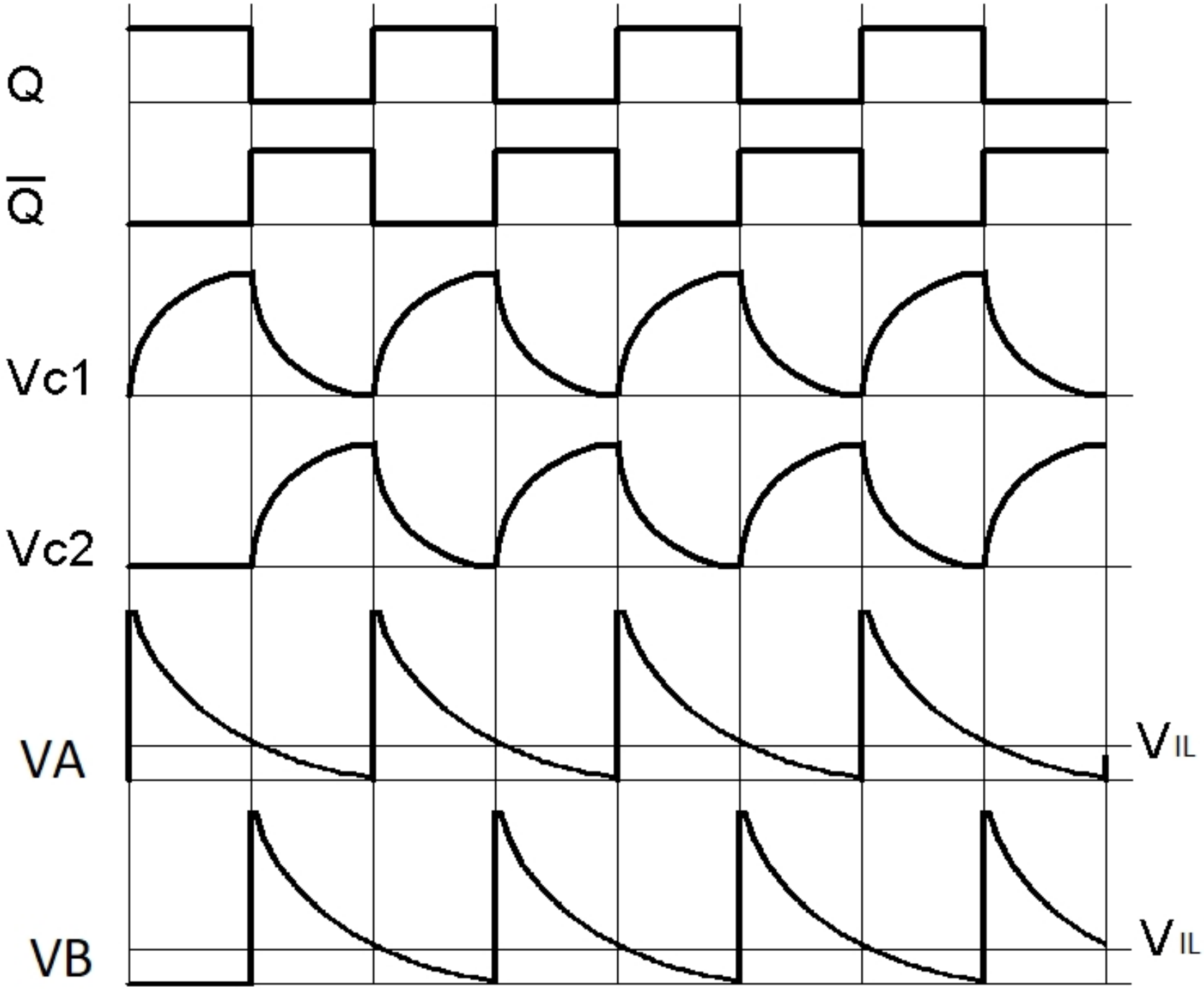
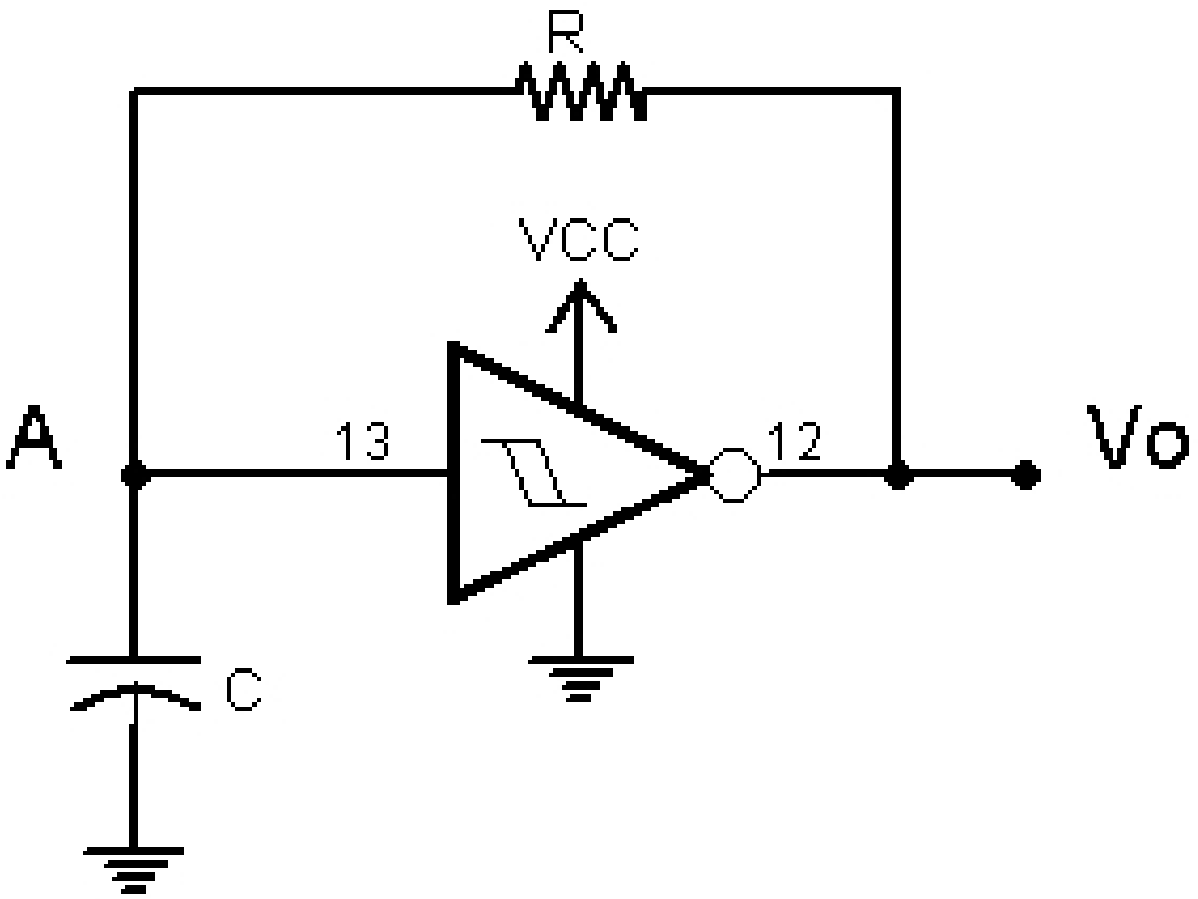




$$T = 1{,}4(R + R_3)C$$

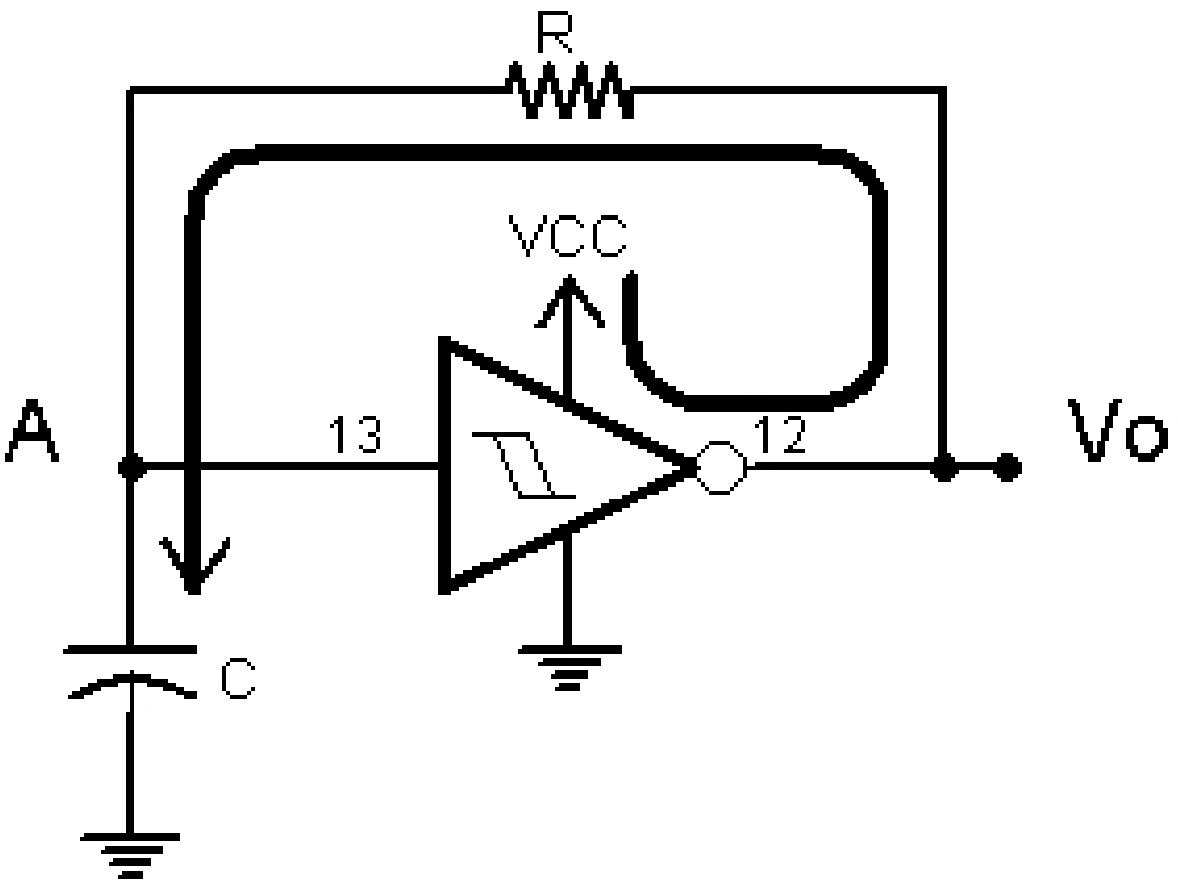
Mạch dao động

- Mạch dao động dùng cổng NOT



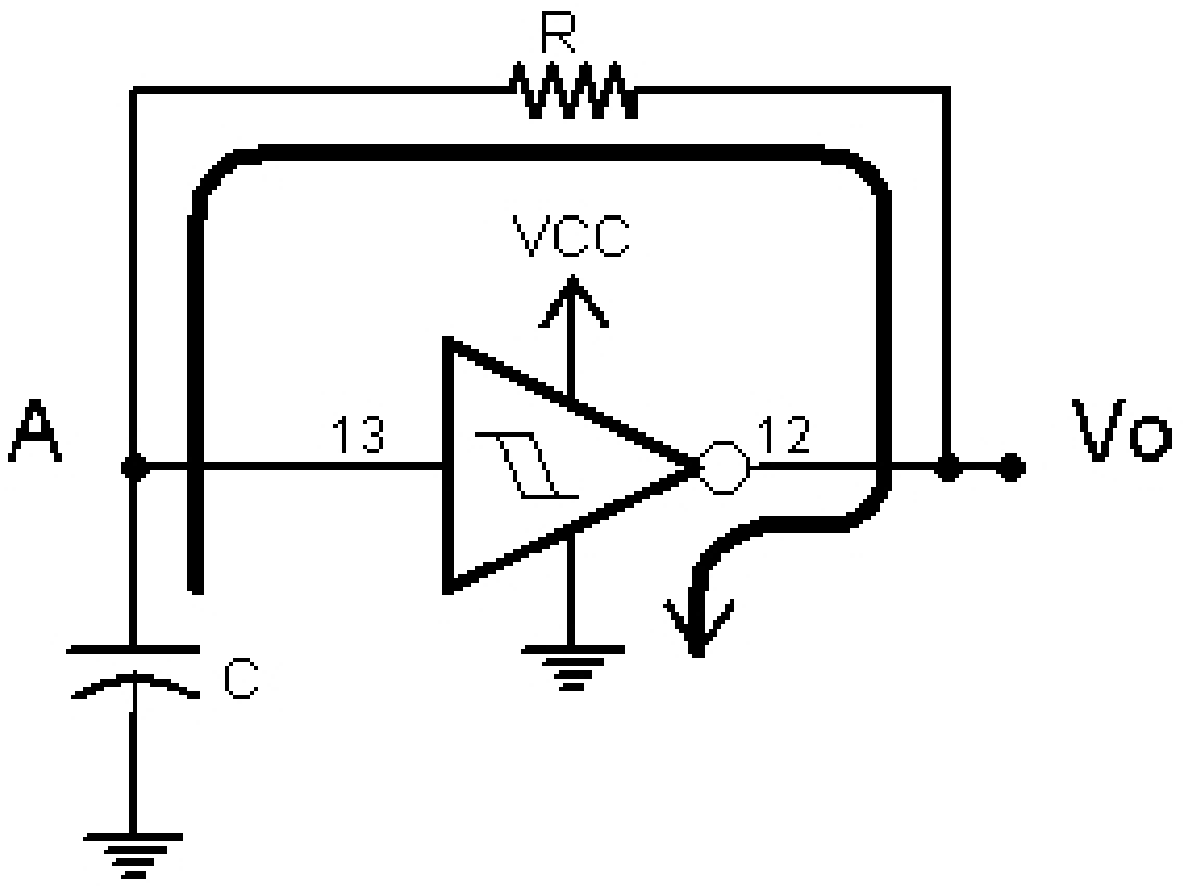
Mạch dao động

- Mạch dao động dùng cổng NOT



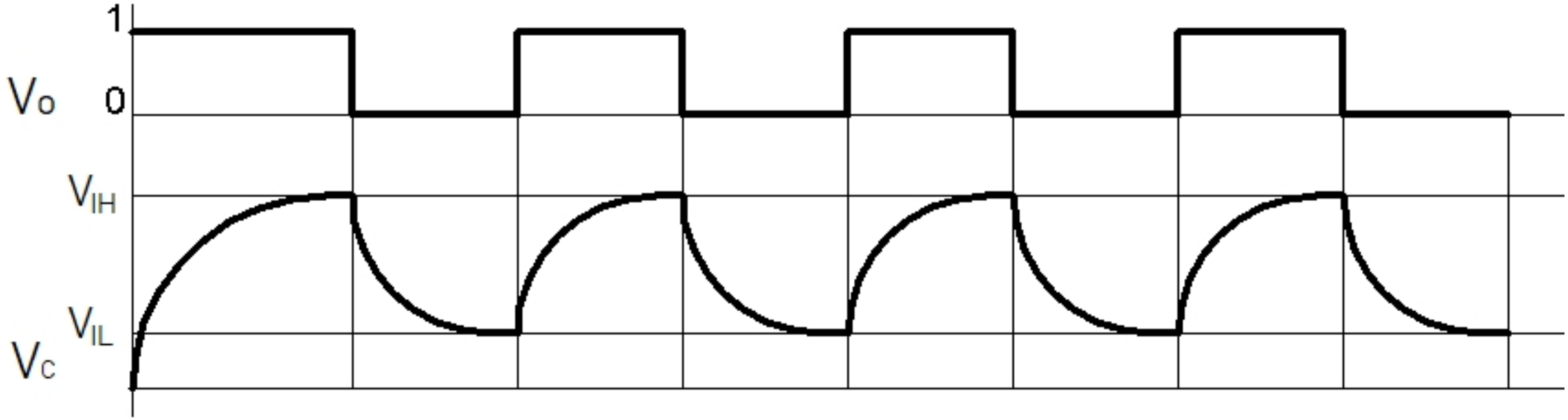
Mạch dao động

- Mạch dao động dùng cổng NOT



Mạch dao động

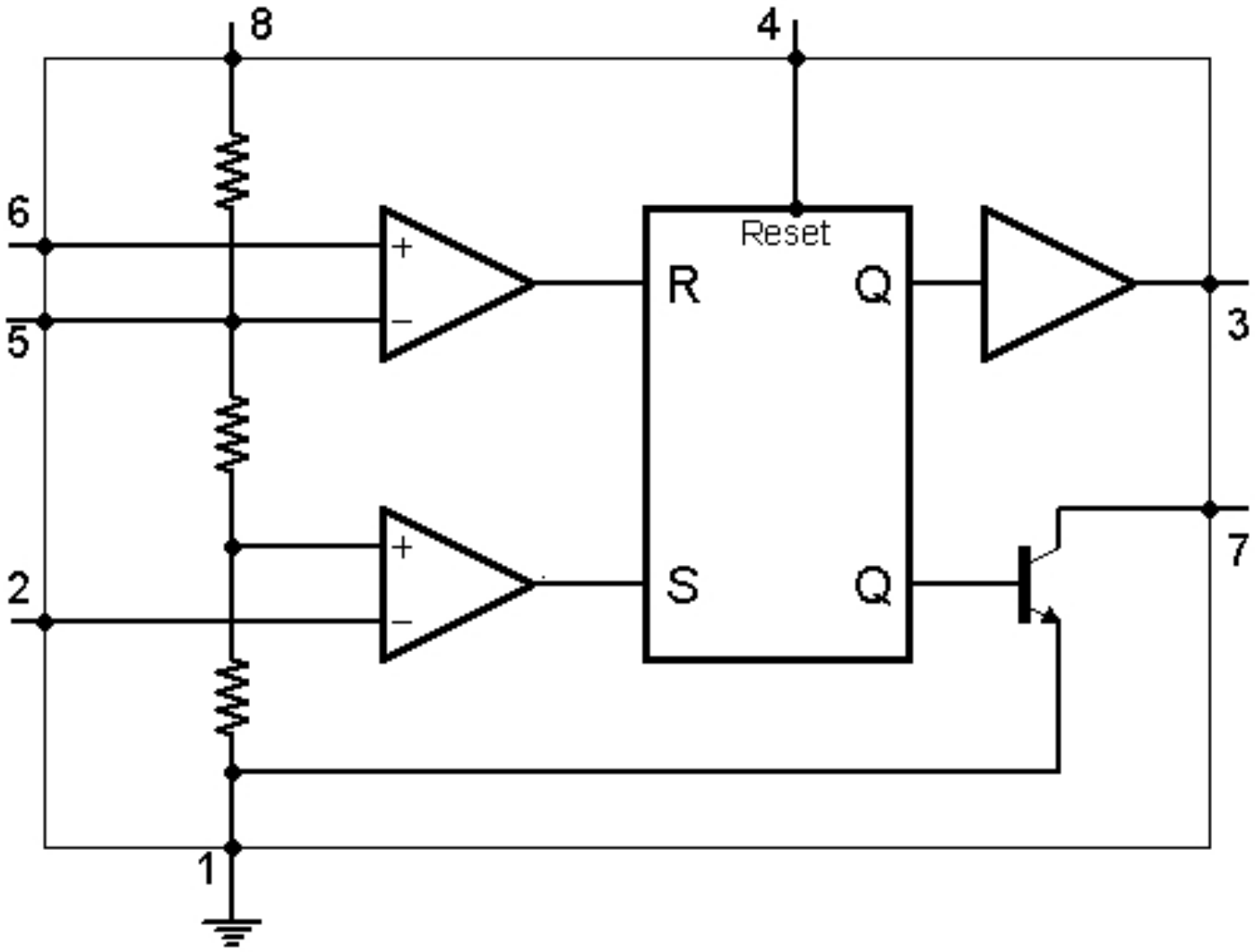
- Mạch dao động dùng cổng NOT



$$T = 2RC\ln\frac{V_{OH} - V_{IL}}{V_{OH} - V_{IH}}$$

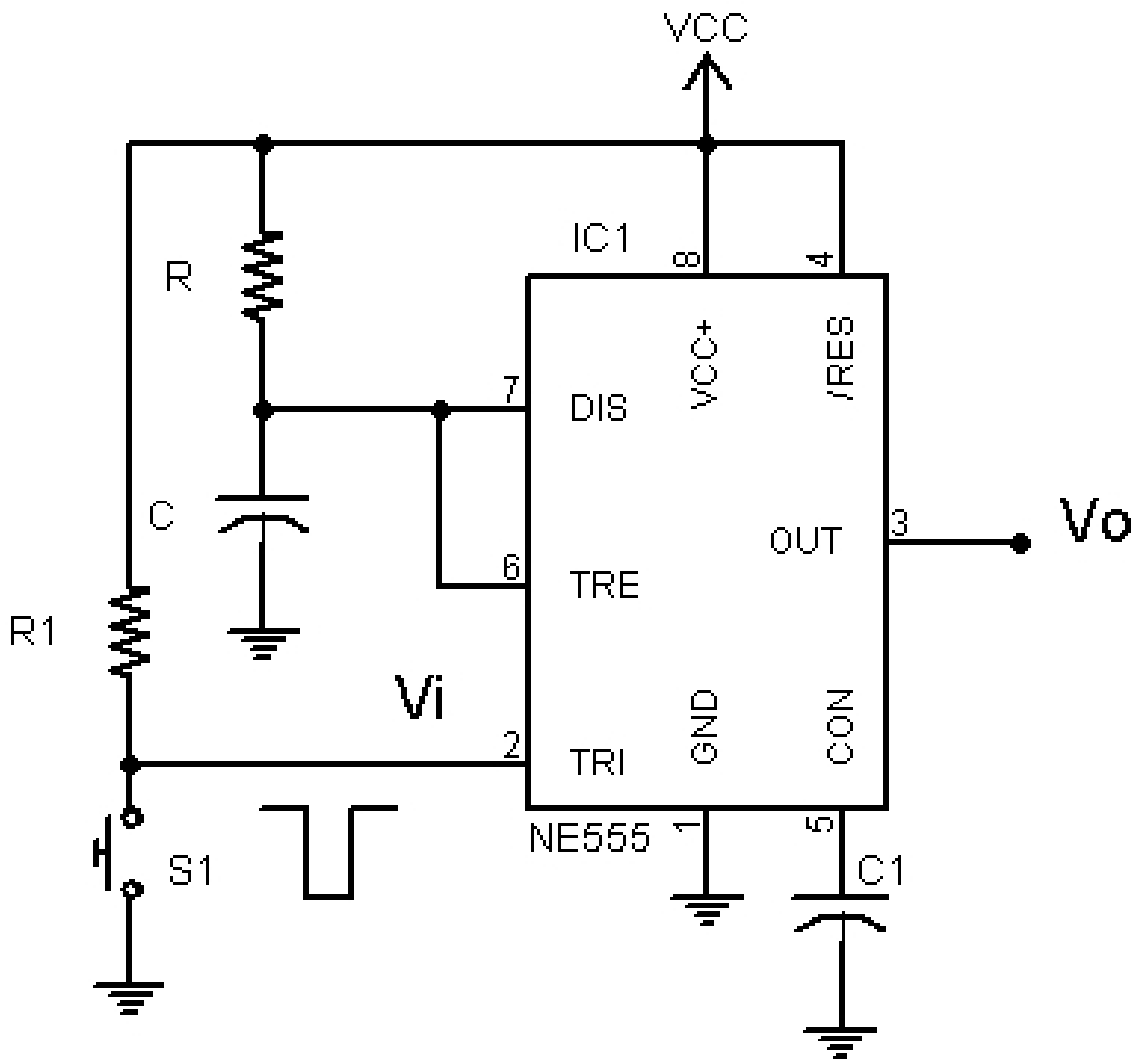
IC 555

- Sơ đồ IC 555



IC 555

- Mạch đơn ổn dùng IC 555



IC 555

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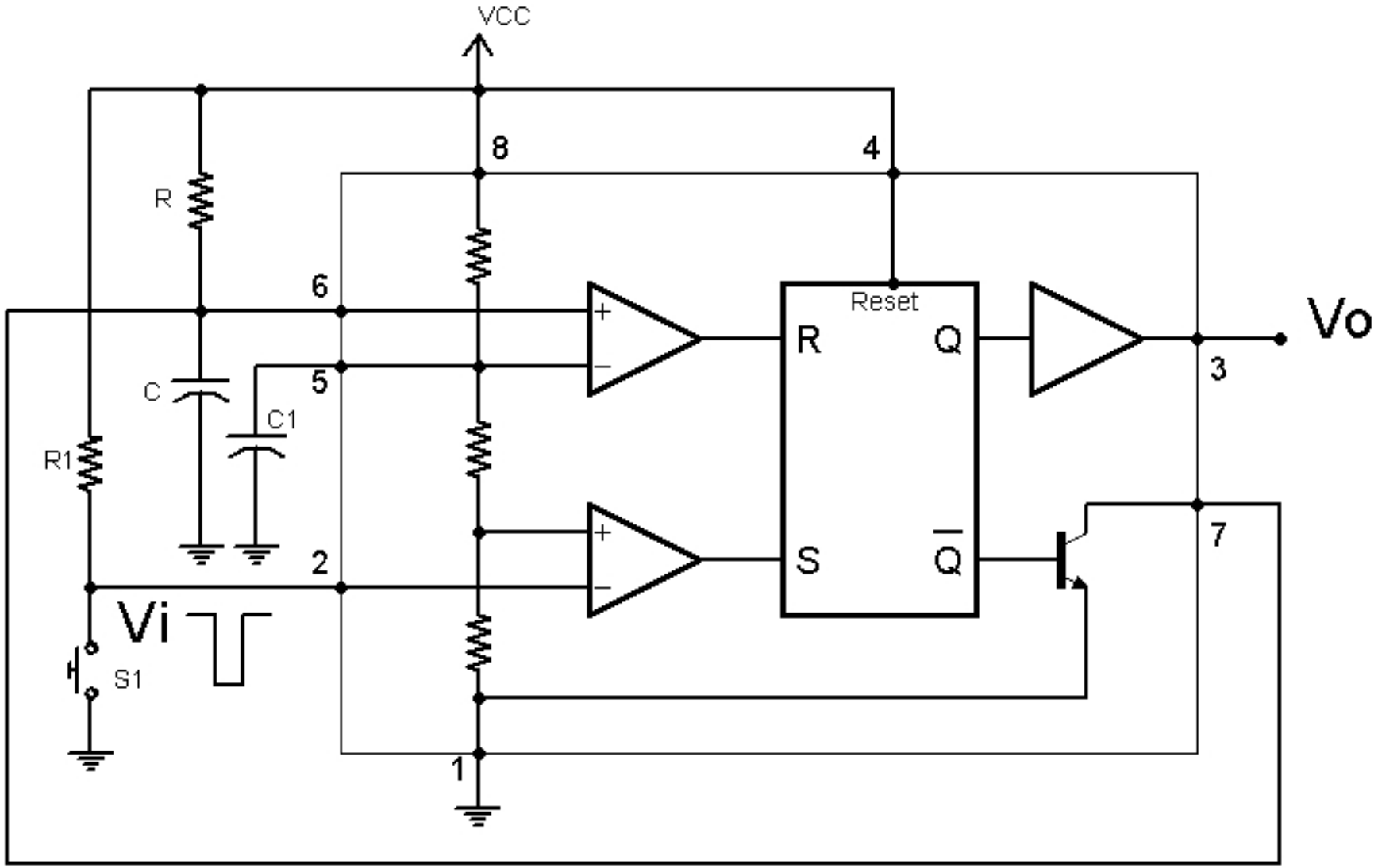

IC 555

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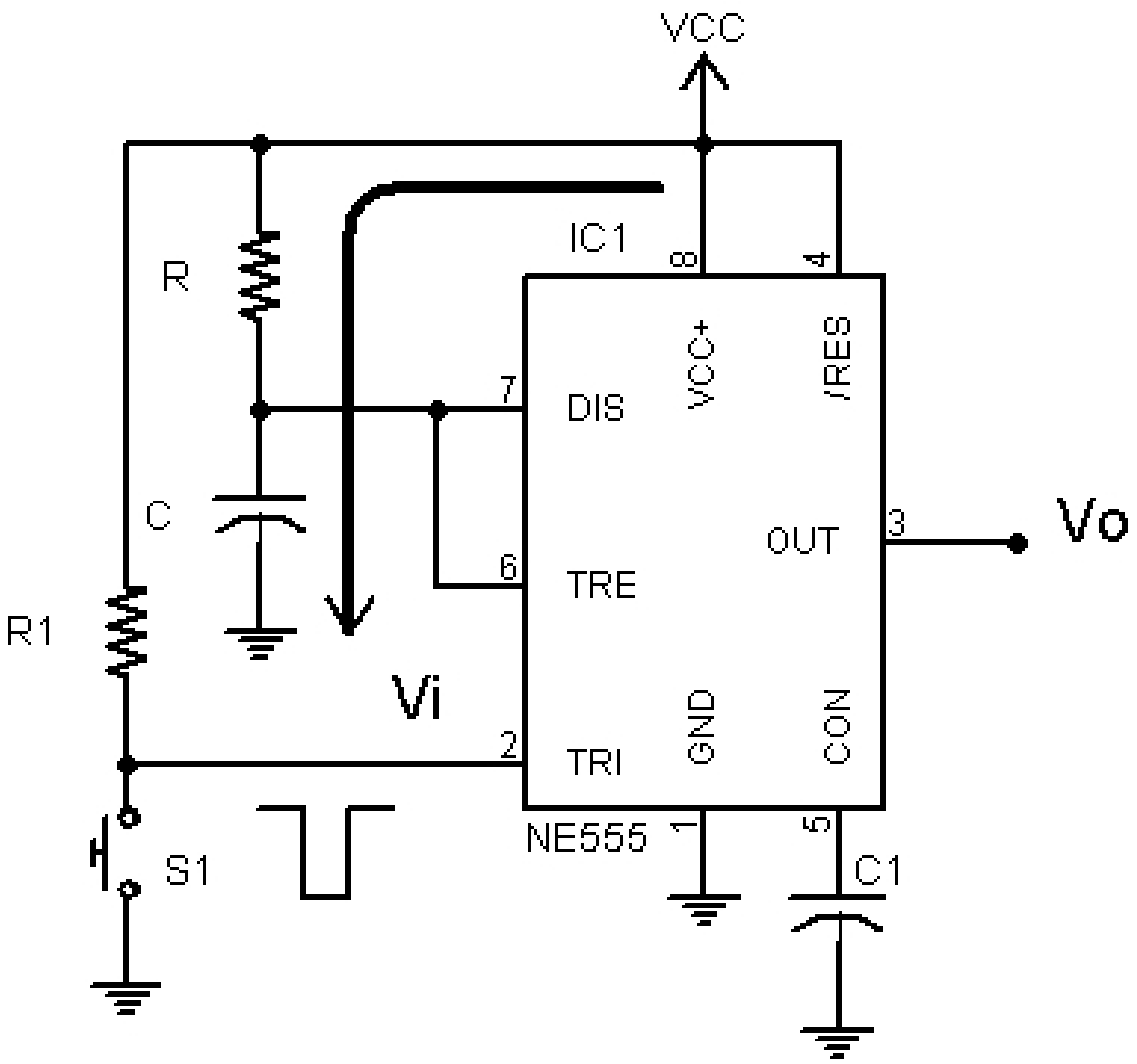
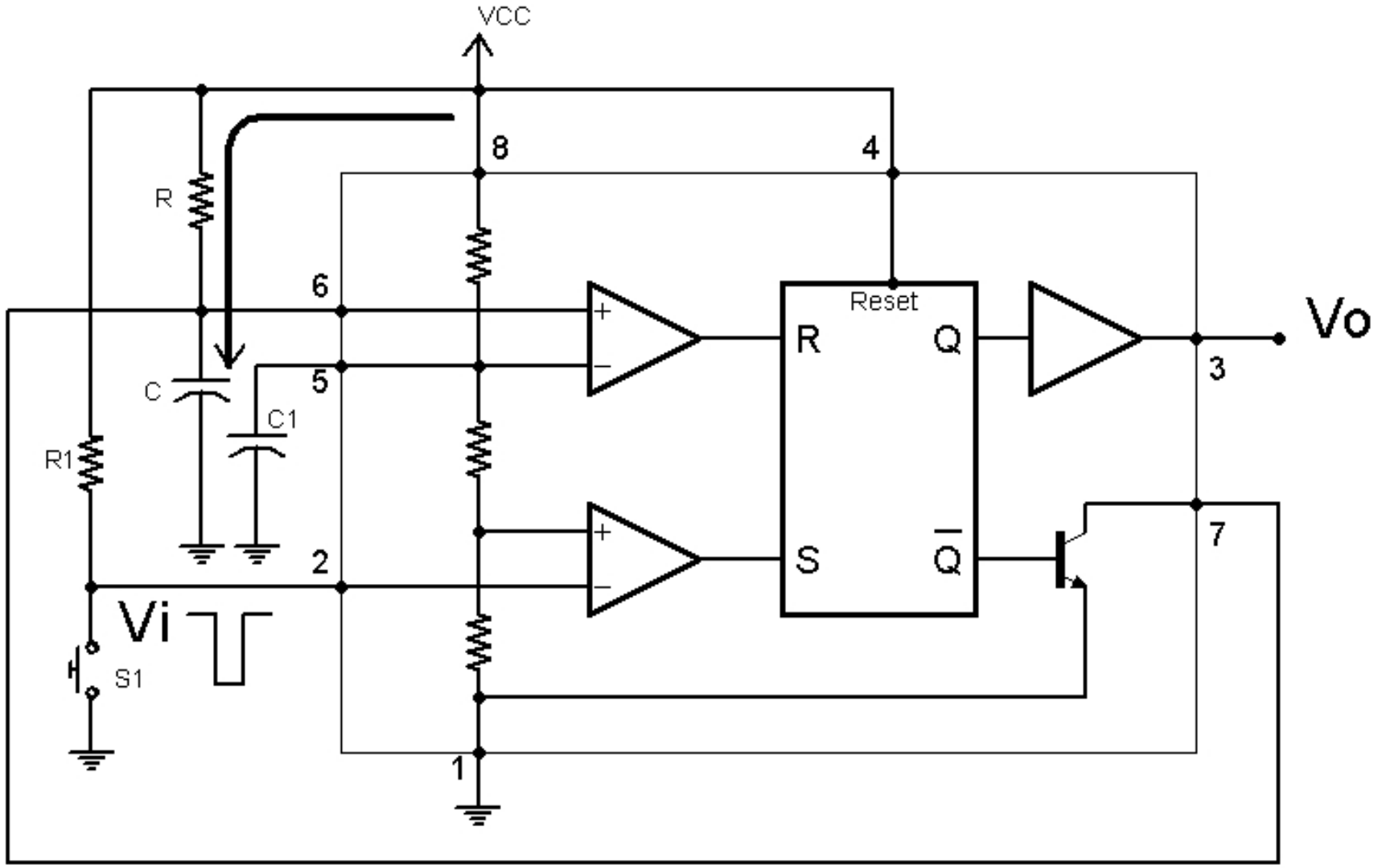


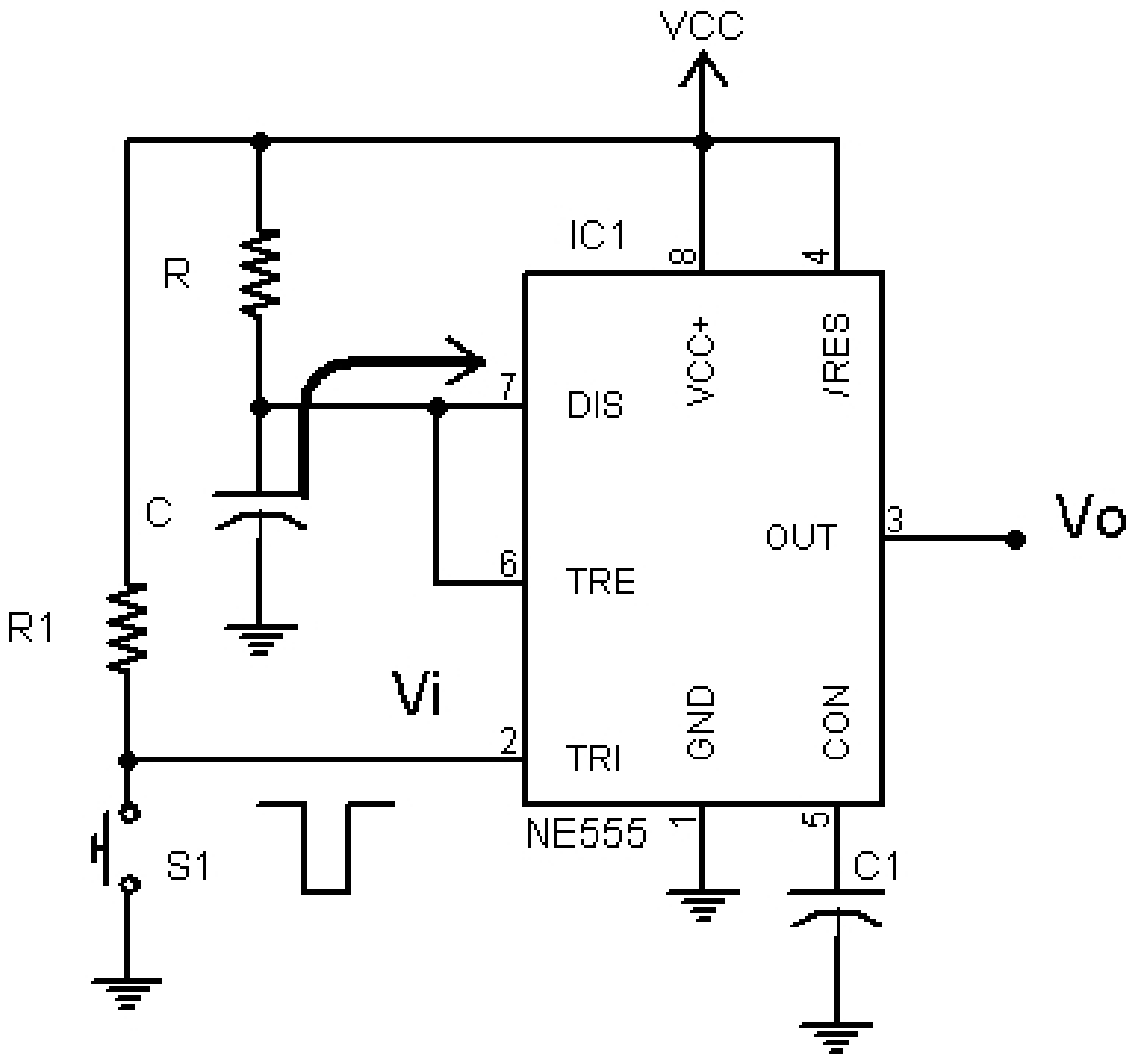
IC 555

- Mạch đơn ổn dùng IC 555



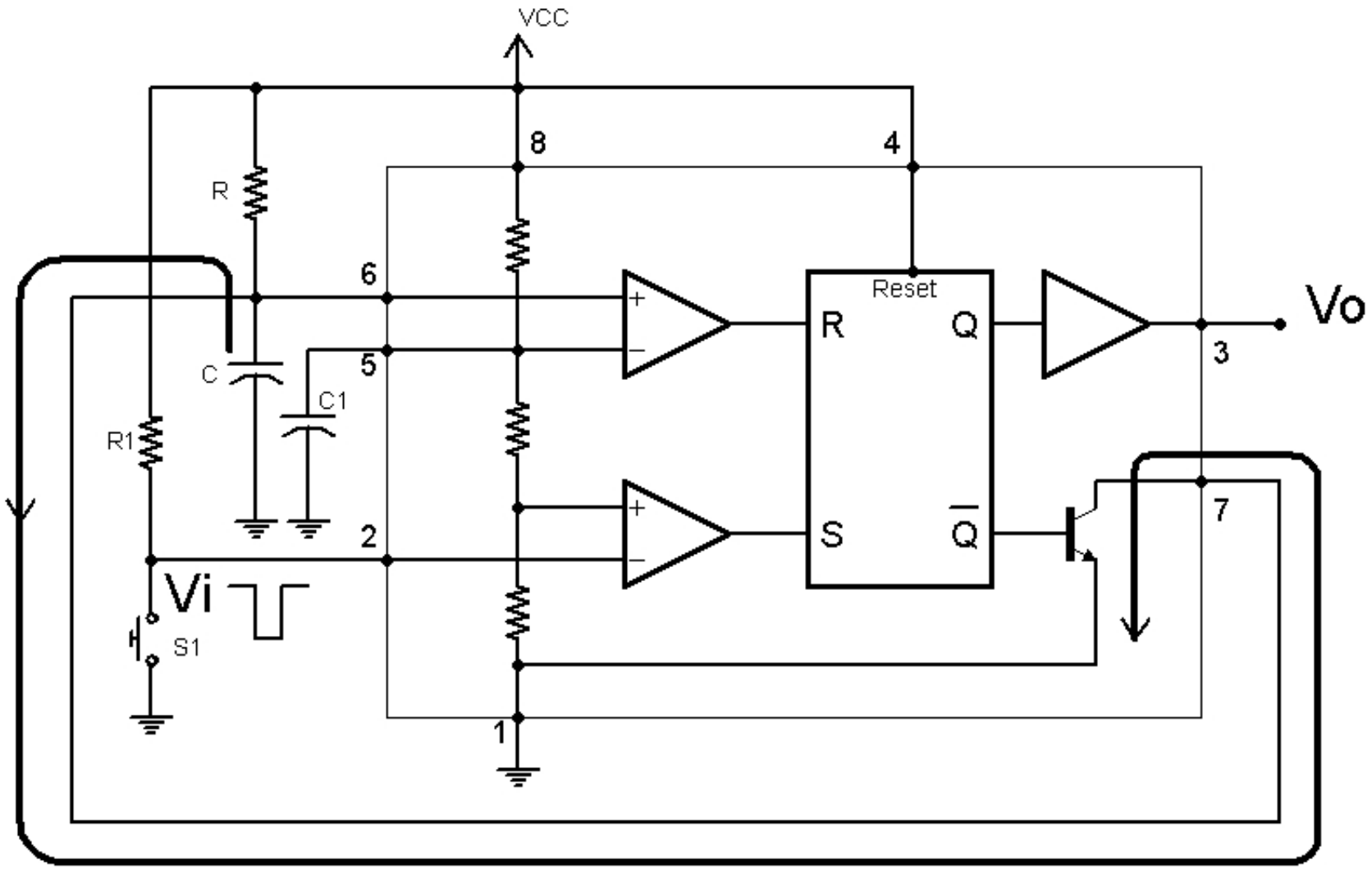
IC 555

- Mạch đơn ổn dùng IC 555



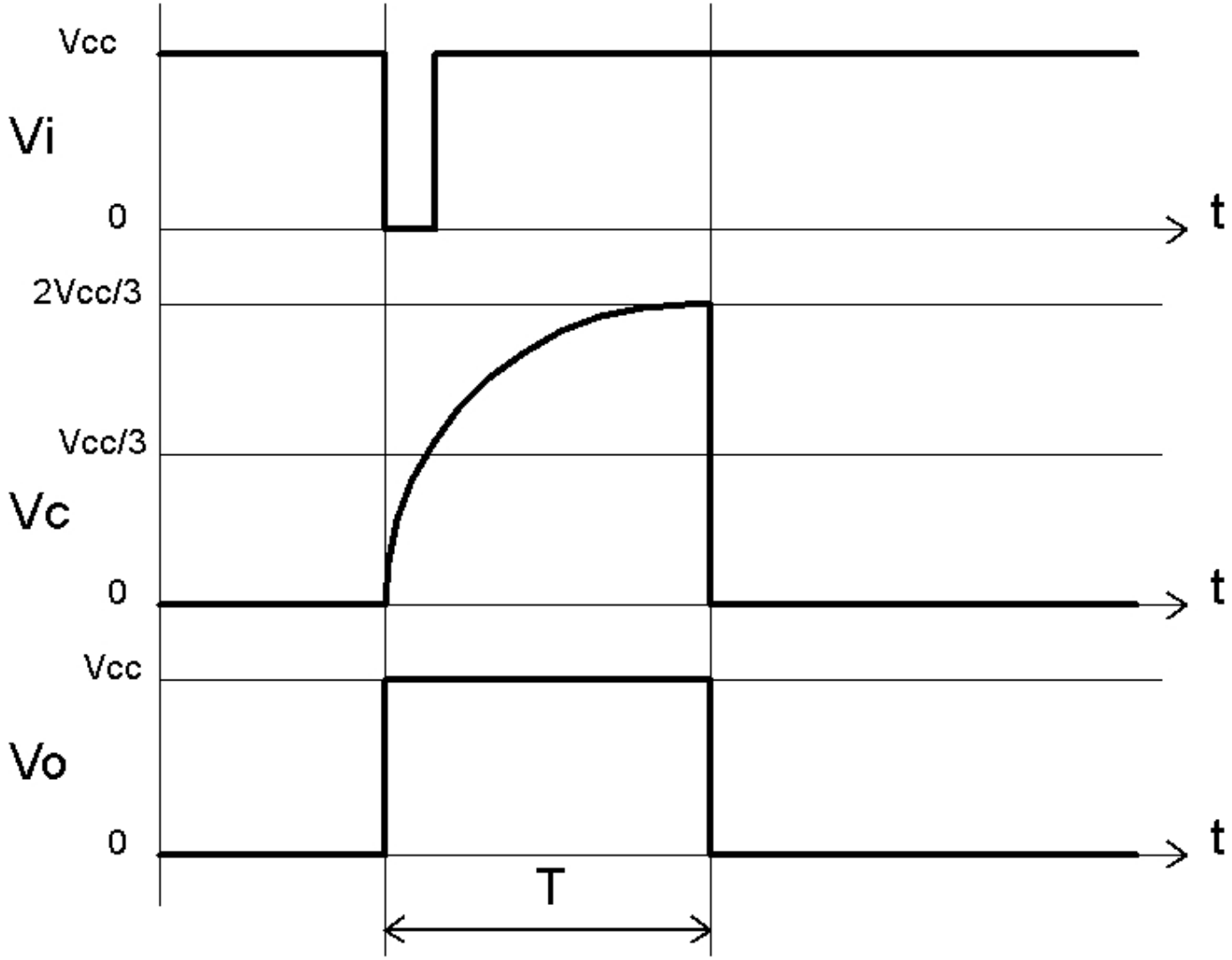
IC 555

- Mạch đơn ổn dùng IC 555



IC 555

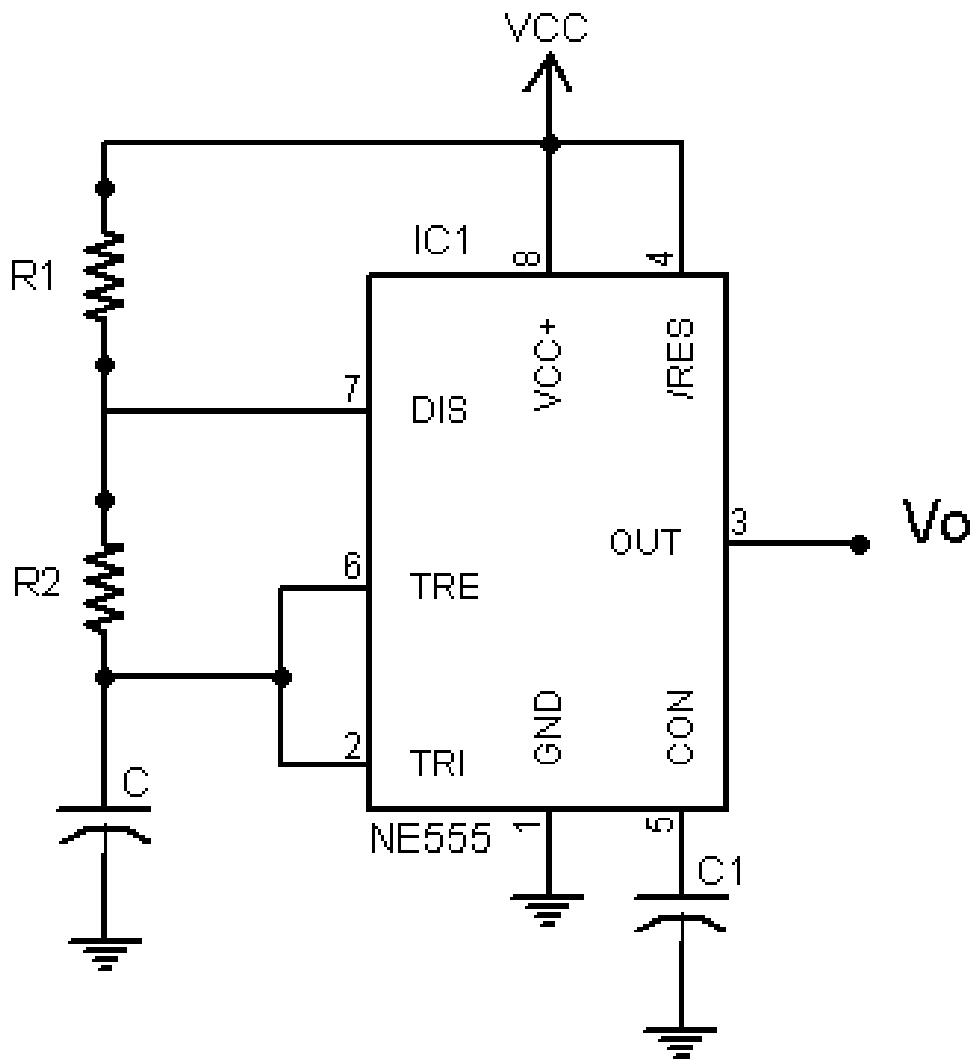
- Mạch đơn ổn dùng IC 555



$$T = RC\ln 3$$

IC 555

- Mạch dao động dùng IC 555



IC 555

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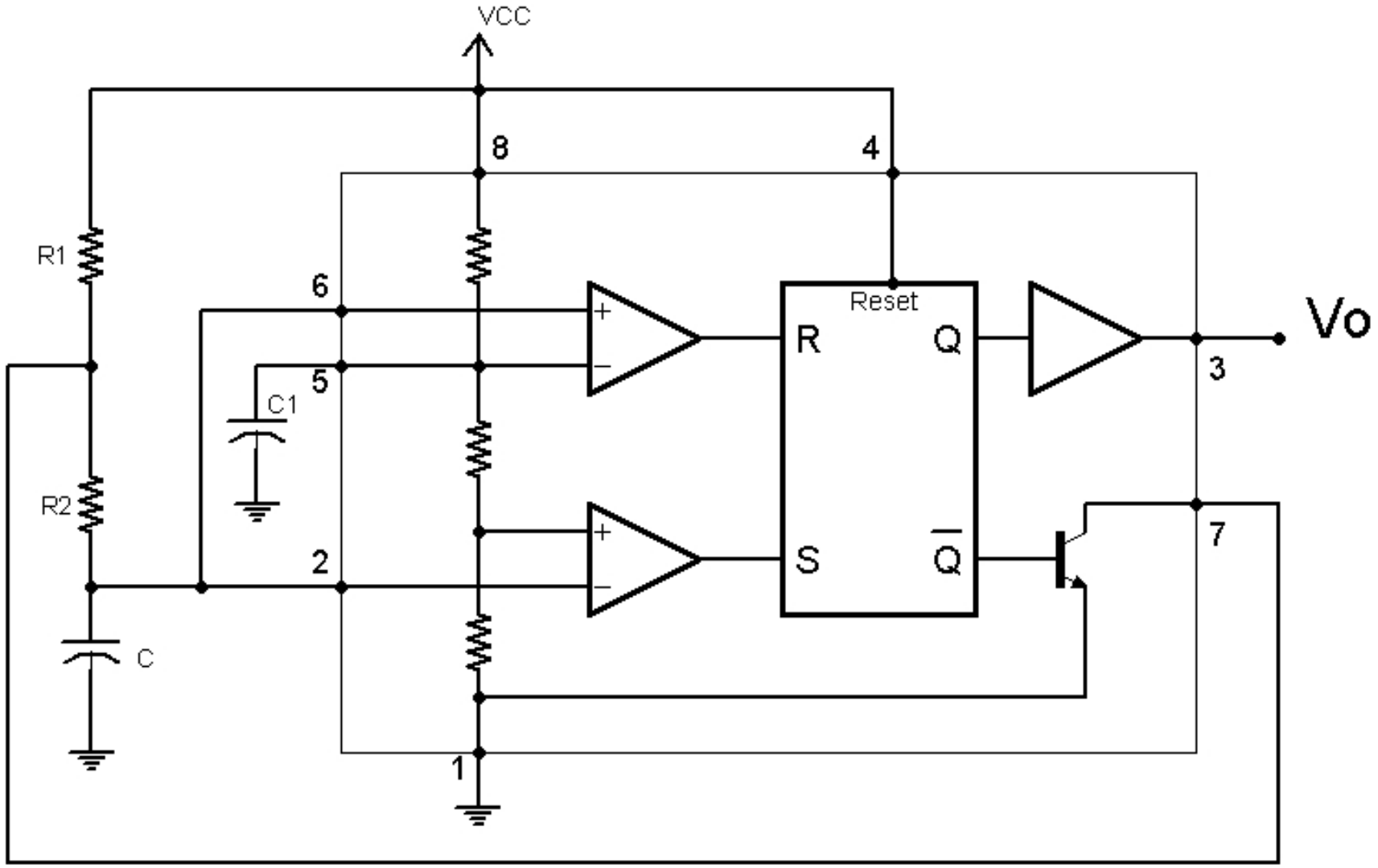
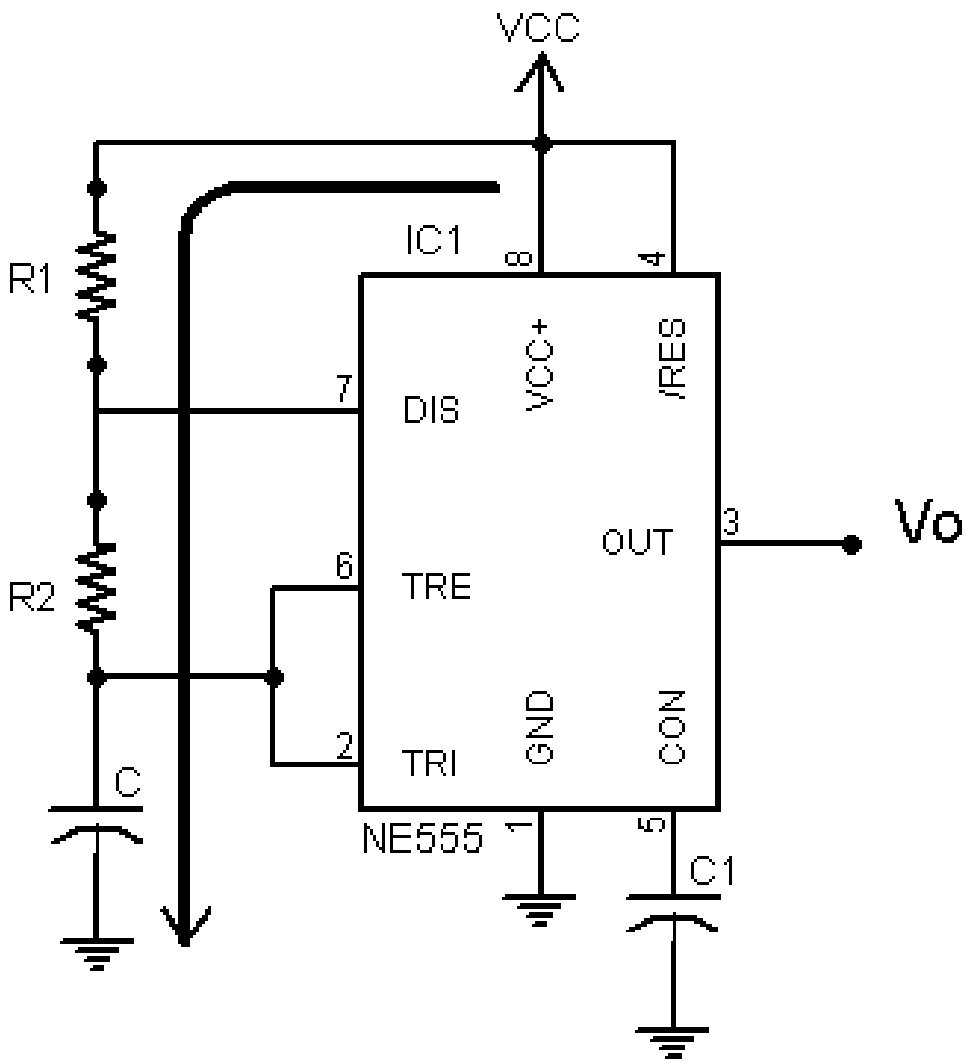


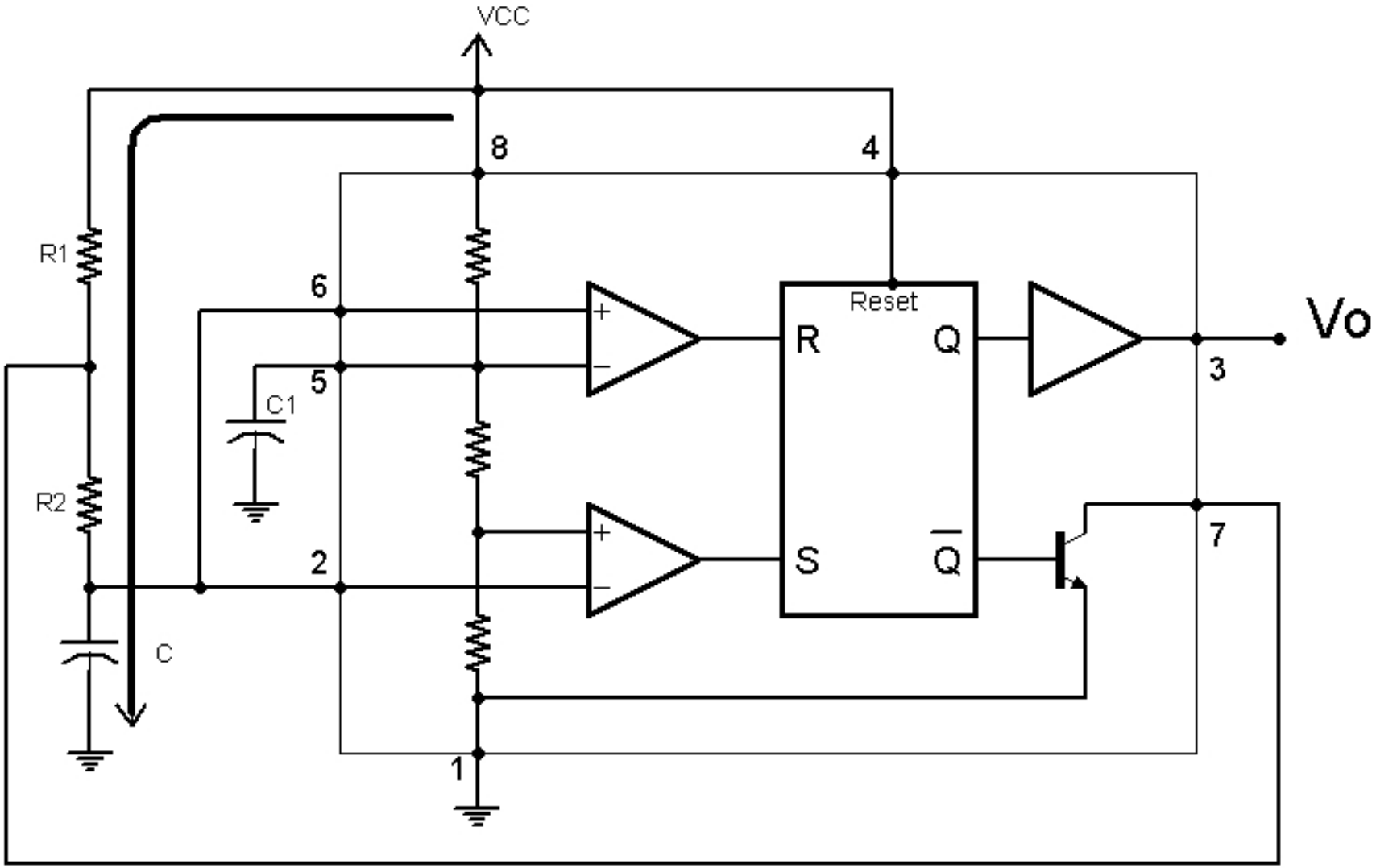
IC 555

- Mạch dao động dùng IC 555



IC 555

- Mạch dao động dùng IC 555



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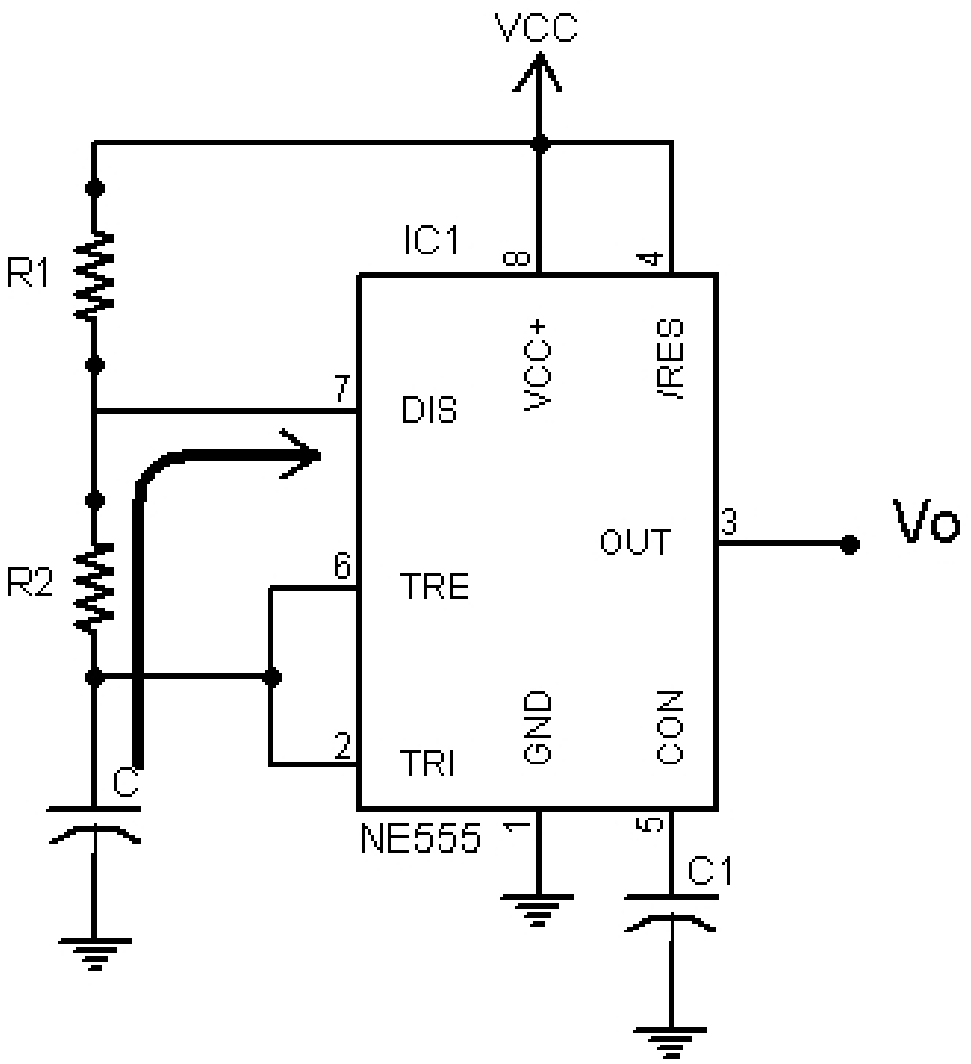

IC 555

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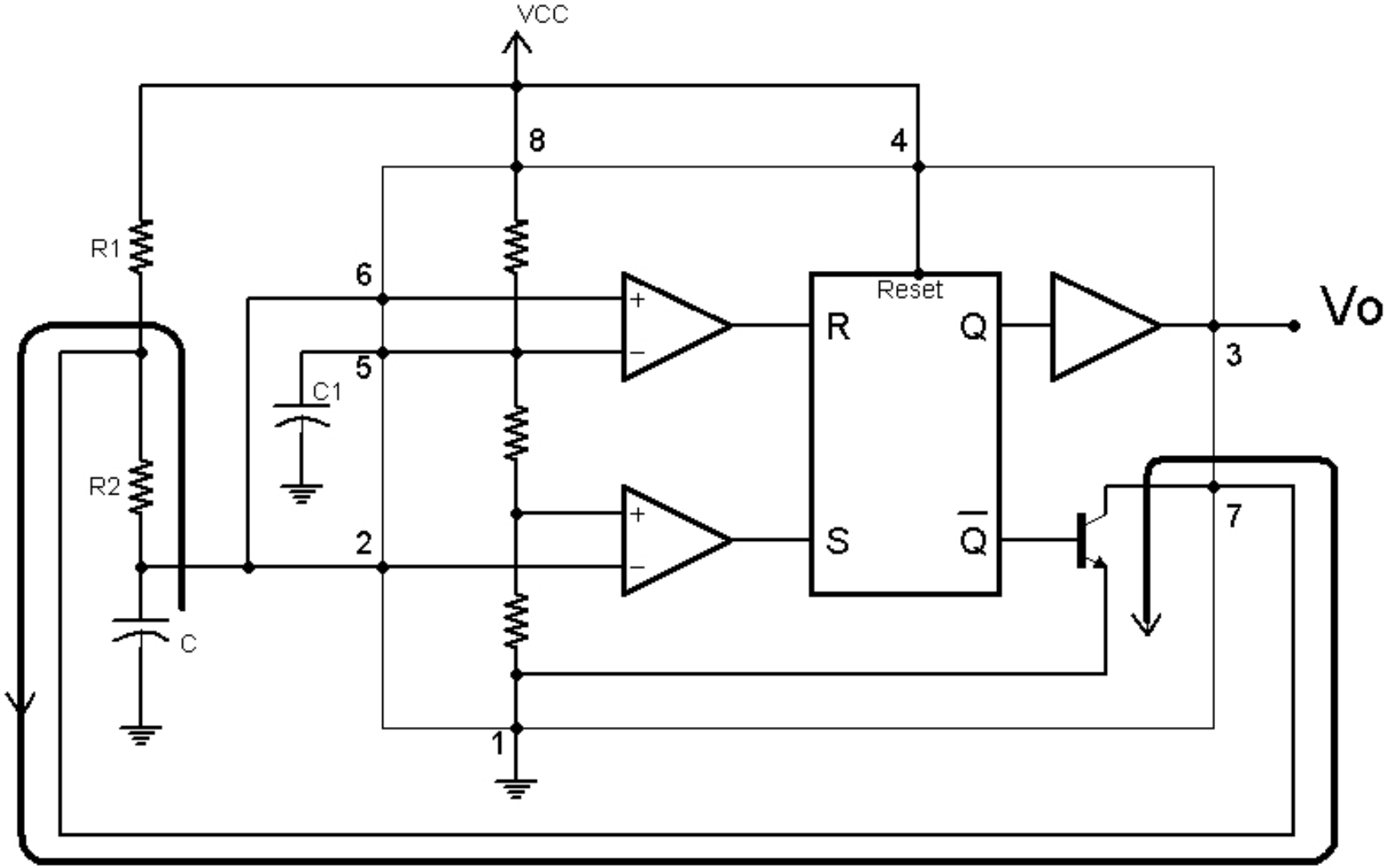

IC 555

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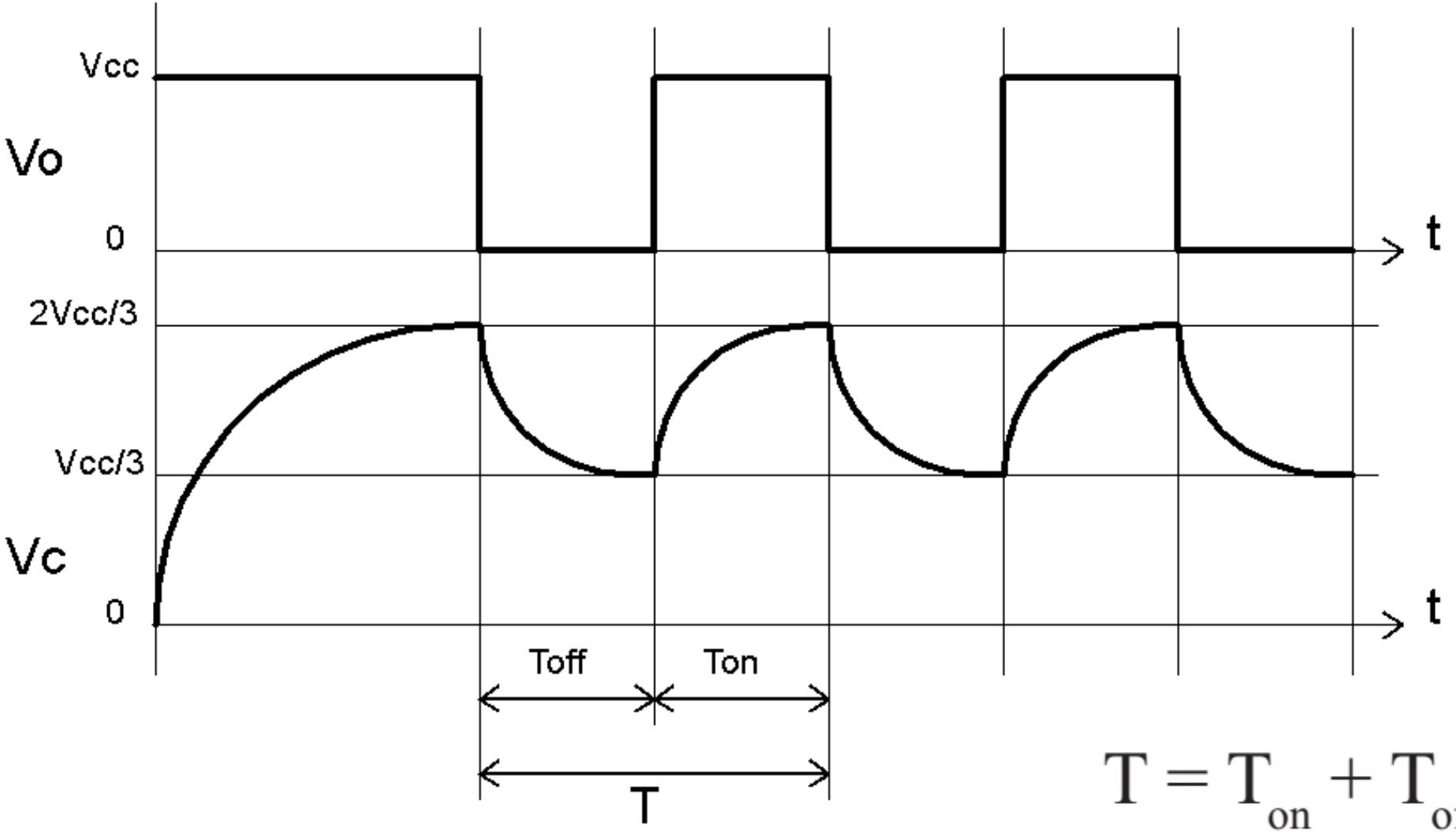
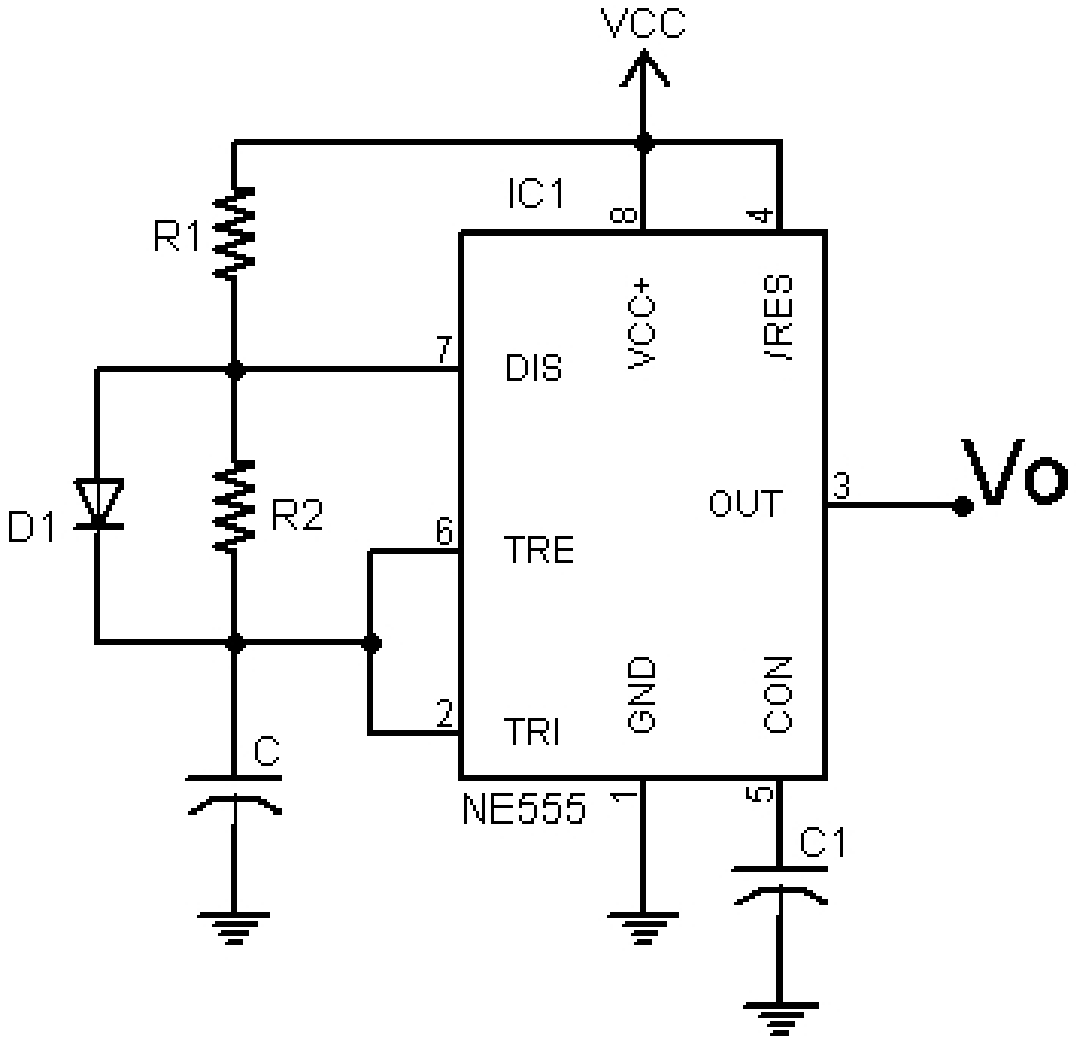


$$T = T_{on} + T_{off} = (R_1 + 2R_2)C\ln 2$$

IC 555

- Mạch dao động dùng IC 555